Special-Purpose Steam Turbines For Refinery Services

API STANDARD 612
SECOND EDITION, JUNE 1979

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Refining Department

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FOREWORD

This standard is based on the accumulated knowledge and experience of petroleum refiners and turbine manufacturers. The object of this publication is to provide a purchase specification to facilitate the manufacture and procurement of special-purpose turbines for refinery service. This standard covers the minimum requirements for special-purpose steam turbines for general refinery systems, including basic design, materials, related lube-oil systems, controls, and auxiliary equipment.

This standard requires the purchaser to specify certain details and features. Also, it is recognized that the purchaser may desire to modify, delete, or amplify sections of the standard. It is strongly recommended that such modifications, deletions, and amplifications be made by supplementing this standard, rather than by rewriting or by incorporating sections into another complete standard.

Suggested revisions are invited and should be submitted to the director, Refining Department, American Petroleum Institute, 2101 L Street, N.W., Washington, D.C. 20037.

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CONTENTS

PAGE

SECTION 1 — GENERAL
1.1 Scope 1
1.2 Alternative Design 1
1.3 Conflicting Requirements 1
1.4 Definition of Terms 1
1.5 Referenced Publications 2

SECTION 2 — BASIC DESIGN
2.1 General 3
2.2 Casings 4
2.3 Casing Appurtenances 5
2.4 Casing Connections 5
2.5 External Forces and Moments 6
2.6 Rotating Elements 6
2.7 Shaft Seals 6
2.8 Dynamics 7
2.9 Bearings and Bearing Housings 9
2.10 Lubrication and Control Oil System 10
2.11 Materials 11
2.12 Nameplates 12

SECTION 3 — ACCESSORIES
3.1 Couplings and Guards 13
3.2 Mounting Plates 13
3.3 Gear Units 14
3.4 Controls, Protective Devices, and Instrumentation 14
3.5 Gland Vacuum System 17
3.6 Piping and Appurtenances 17
3.7 Insulation and Jacketing 17
3.8 Weather Protection 17
3.9 Special Tools 17
3.10 Turning Gear 17

SECTION 4 — INSPECTION AND TESTING
4.1 General 18
4.2 Inspection 18
4.3 Testing 18
4.4 Preparation for Shipment 20

SECTION 5 — GUARANTEE AND WARRANTY
5.1 Mechanical 21
5.2 Performance 21

SECTION 6 — VENDOR'S DATA
6.1 Proposals 22
6.2 Contract Data 22

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APPENDIX A — GLAND SEALING AND LEAK-OFF EQUIPMENT 27

APPENDIX B — RATED LOAD CURVES FOR STANDARD 6 x 6 SHOE THRUST BEARINGS 29

APPENDIX C — TYPICAL DATA SHEETS 33

APPENDIX D — VENDOR DRAWING AND DATA REQUIREMENTS 43

LIST OF ILLUSTRATIONS

Figures

1 — Rotor Response Plot 8
2 — Balancing Machine Response Check 9
B-1 — Rated Load Versus Shaft Speed 31
B-2 — Rated Load Versus Shaft Speed 32

Tables

1 — Instrument Gages 15
2 — Alarms and Shutdowns 15

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Special-Purpose Steam Turbines for Refinery Services

SECTION 1 — GENERAL

1.1 Scope

1.1.1 This standard covers the minimum requirements for special-purpose steam turbines for general refinery services. These include basic design, materials, related lube-oil systems, controls, and auxiliary equipment.

NOTE: A bullet (•) in the margin indicates where a decision by the purchaser is required. These decisions should be indicated directly on the data sheets when provisions are made for them; otherwise, they should be stated in the quotation request or in the order.

• **1.1.2** Steam turbines, that is, general and special-purpose, are classified according to service requirements. The purchaser shall specify the type of turbine dependent upon his needs. This requirement shall be indicated on the data sheet.

1.1.2.1 General-purpose turbines are those horizontal or vertical turbines used to drive equipment that usually is spared, is relatively small in size (horsepower), or is in noncritical service. They are intended for applications where steam conditions will not exceed 600 pounds per square inch gage (41 bar) pressure or 750F (400C) inlet temperature or both, and where speed will not exceed 6000 revolutions per minute.

Requirements for general-purpose turbines are defined in API Standard 611.

1.1.2.2 Special-purpose turbines are those horizontal turbines used to drive equipment that usually is not spared, is relatively large in size (horsepower), or is in critical service. This category is not limited by steam conditions or turbine speed.

1.2 Alternative Design

1.2.1 The vendor may offer an alternative design (see 6.1.1, item 9). Equivalent metric dimensions, fasteners, and flanges may be substituted as mutually agreed between purchaser and vendor.

1.3 Conflicting Requirements

1.3.1 In case of conflict between this standard and the inquiry or order, the information included in the order shall govern.

1.4 Definition of Terms

The terms used in this standard are defined as follows:

Normal applies to the horsepower, speed, and steam conditions expected in usual operation and to the point at which the highest efficiency is desired.

Rated is the greatest turbine horsepower specified and corresponding speed. It includes all the margin required by the driven equipment specifications.

Maximum continuous speed is the upper limit of the operating speed of the turbine. It shall be at least 105 percent of the highest operating speed required by the driven machine to meet the specified operating conditions.

Maximum allowable speed is the highest speed at which the manufacturer's design will permit continuous operation, assuming the proper overspeed and governor mechanisms are installed.

Trip speed is the speed at which the independent, emergency overspeed device operates to shut-off steam to the turbine. It shall be approximately 110 percent of the maximum continuous speed.

Maximum inlet refers to the highest inlet steam pressure and temperature conditions at which the turbine is required to operate continuously.

Minimum inlet refers to the lowest inlet steam pressure and temperature conditions at which the turbine is required to operate continuously.

Maximum exhaust is the highest exhaust steam pressure at which the turbine is required to operate continuously.

Minimum exhaust is the lowest exhaust steam pressure at which the turbine is required to operate continuously.

Maximum exhaust casing pressure is the highest exhaust steam pressure which the purchaser requires the exhaust casing to contain with steam supplied at maximum inlet conditions. It is the exhaust pressure that exists at full flow of the relief valve.

Maximum allowable working pressure is the maximum continuous pressure for which the manufacturer has designed the equipment (or any part to which the term is

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referred) when handling the specified fluid at the specified temperature.

NEMA rated inlet and exhaust are equivalent to data sheet maximum inlet and exhaust steam conditions.

Axial split refers to casing joints which are parallel to the shaft centerline.

Radial split refers to casing joints which are transverse to the shaft centerline.

Critical or resonant speed—see 2.8.

Use of the word *design* or *mechanical design* in any term such as design horsepower, design pressure, design temperature, or design speed should be avoided in the purchaser's specifications. This terminology should be used only by the equipment designer or manufacturer.

1.5 Referenced Publications

1.5.1 The latest edition of the following standards, codes, or specifications shall, to the extent specified, form a part of this standard.

ANSI [1]

B1.1	*Unified Inch Screw Threads (UN and UNR Thread Forms)*
B2.1	*Pipe Threads (Except Dryseal)*
B16.1	*Cast Iron Pipe Flanges and Flanged Fittings, Class 25, 125, 250, and 800*
B16.5	*Steel Pipe Flanges and Flanged Fittings*
B16.11	*Forged Steel Fittings, Socket-Welding and Threaded*
B17.1	*Keys and Keyseats*
B31.3	*Code for Chemical Plant and Petroleum Refinery Piping*

API

Std 605	*Large-Diameter Carbon Steel Flanges*
Std 611	*General-Purpose Steam Turbines for Refinery Services*
Std 613	*Special-Purpose Gear Units for Refinery Services*
Std 614	*Lubrication, Shaft Sealing, and Control Oil Systems for Special-Purpose Applications*
Std 615	*Sound Control of Mechanical Equipment for Refinery Services*
Std 670	*Noncontacting Vibration and Axial Position Monitoring Systems*
Std 671	"Special-Purpose Couplings for General Refinery Services" (in preparation)
RP 550	*Manual on Installation of Refinery Instruments and Control Systems*

ASME [2]

Boiler and Pressure Vessel Code:	Section VIII, Pressure Vessels, and Section IX, Welding and Brazing Qualifications
PTC 6	*Performance Test Code*, "Steam Turbines"

ASTM [3]

A 53	*Pipe, Steel, Black and Hot-Dipped, Zinc-Coated Welded and Seamless*
A 106	*Seamless Carbon Steel Pipe for High-Temperature Service*
A 192	*Seamless Carbon Steel Boiler Tubes for High-Pressure Service*
A 193	*Alloy Steel and Stainless Steel Bolting Materials for High-Temperature Service*
A 194	*Carbon and Alloy Steel Nuts for Bolts for High-Pressure and High-Temperature Service*
A 216	*Carbon Steel Castings Suitable for Fusion Welding for High-Temperature Service*
A 269	*Seamless and Welded Austenitic Stainless Steel Tubing for General Service*
A 278	*Gray Iron Castings for Pressure-Containing Parts for Temperatures up to 650 F*
A 307	*Carbon Steel Externally and Internally Threaded Standard Fasteners*
A 325	*High-Strength Bolts for Structural Steel Joints Including Suitable Nuts and Plain Hardened Washers*
A 395	*Ferritic Ductile Iron Pressure-Retaining Castings for Use at Elevated Temperatures*
A 418	*Ultrasonic Inspection of Turbine and Generator Steel Rotor Forgings*
A 472	*Tests for Heat Stability of Steam Turbine Shafts and Rotor Forgings*
A 536	*Ductile Iron Castings*
A 668	*Steel Forgings, Carbon and Alloy, for General Industrial Use*
E 94	*Recommended Practice for Radiographic Testing*
E 109	*Dry Powder Magnetic Particle Inspection*
E 125	*Reference Photographs for Magnetic Particle Indications on Ferrous Castings*
E 138	*Wet Magnetic Particle Inspection*
E 142	*Controlling Quality of Radiographic Testing*

MSS [4]

SP44	*Steel Pipeline Flanges*

[1] American National Standards Institute, 1430 Broadway, New York, N.Y. 10018.

[2] American Society of Mechanical Engineers, 345 East 47th Street, New York, N.Y. 10017.

[3] American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103.

[4] Manufacturers Standardization Society of the Valve and Fittings Industry, 1815 North Fort Myer Drive, Arlington, Va. 22209.

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NEMA [5]
Proposed 1979 publication: "Steam Turbines for Mechanical Drive Service"

NFPA [6]
Bulletin No. 70 *National Electrical Code*, Chapter 5, Articles 500, "Hazardous (Classified) Locations," and 501, "Class I Locations"

TEMA [7]
Standards, Codes, and Specifications

OSHA [8]
Occupational Safety and Health Standards of the U.S. Department of Labor

• **1.5.2** The applicability of federal, state, or local codes, regulations, ordinances, or rules shall be mutually agreed upon by the purchaser and vendor.

SECTION 2 — BASIC DESIGN

2.1 General

2.1.1 The equipment, including auxiliaries, covered by this standard shall be suitable for the specified operating conditions and shall be designed and constructed for at least 3 years of continuous service. It is recognized this is a design criterion and that uninterrupted operation for this period of time involves factors beyond the vendor's control.

2.1.2 Turbines shall be capable of the following:

1. Operation at normal power and speed with normal steam conditions. Steam rate shall be guaranteed at these conditions.

• 2. Delivering rated load at its corresponding speed with coincident minimum inlet and maximum exhaust conditions.

NOTE: For noncondensing turbines it may be desirable to specify normal load or some other load instead of rated load to limit maximum turbine capability to a selected percentage of normal horsepower at normal steam conditions in order to obtain higher operating efficiency. For condensing turbines with large condition margins, similar conditions may apply.

3. Continuous operation at maximum continuous speed and at any speed within the range specified.

4. Continuous operation at rated power and speed with maximum inlet steam conditions and maximum or minimum exhaust steam conditions.

• 5. Continuous operation at the lowest speed at which maximum torque is required with minimum inlet and maximum exhaust conditions. The purchaser will specify both the speed and torque values required.

6. Continuous operation at conditions agreed upon between purchaser and vendor for extraction or induction, or both.

7. Operation with variations from rated steam conditions per NEMA proposed 1979 publication, "Steam Turbines for Mechanical Drive Service."

8. Operation uncoupled with maximum inlet steam conditions. (Governing instability may occur and require action such as throttling of inlet pressure.)

2.1.3 All equipment shall be designed to run safely to the trip speed setting and to the relief valve setting.

2.1.4 The combined performance of the turbine (and its driven equipment) after installation shall be the joint responsibility of the vendor and the purchaser. The units shall perform substantially as well on their permanent foundation as they did on the vendor's test stand. Many factors (such as piping loads, alignment at operating conditions, supporting structure, handling during shipment, and handling and assembly at site) may adversely affect site performance. In order to minimize the influence of these factors, the vendor shall review and comment on the purchaser's piping and foundation drawings and the vendor's erection supervisor shall observe a check on pip-
• ing alignment by parting the flanges. The supervisor shall observe a turbine alignment check at operating temperature, and when specified, shall be present during the initial alignment check.

2.1.5 The arrangement of the equipment, including piping and auxiliaries, shall be developed jointly by the purchaser and vendor. The arrangement shall provide adequate clearance areas and safe access for operation and maintenance.

2.1.6 All equipment shall be designed to permit rapid and economical maintenance. Major parts such as casing

[5] National Electrical Manufacturers Association, 2101 L Street N.W., Washington, D.C. 20037.

[6] National Fire Protection Association, 470 Atlantic Avenue, Boston, Mass. 02210.

[7] Tubular Exchanger Manufacturers Association, Inc., 331 Madison Avenue, New York, N.Y. 10017.

[8] Occupational Safety and Health Administration, U.S. Department of Labor, Washington, D.C. 20210.

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components and bearing housings shall be designed (shouldered or doweled) to ensure accurate alignment on reassembly.

2.1.7 Oil reservoirs, housings which enclose moving lubricated parts, shaft seals, highly polished parts, instrument and control elements, and so forth, shall be designed to minimize the entrance of moisture, dust, and other foreign matter during periods of operation and idleness.

• **2.1.8** Unless otherwise specified, cooling water systems shall be designed for the following conditions:

Maximum allowable working pressure	75 psig		(5 bar)
Hydrotest pressure	115 psig		(7.7 bar)
Maximum pressure drop	15 psig		(1 bar)
Maximum inlet temperature	90 F		(32 C)
Maximum temperature rise	30 F	or	17 C
Fouling factor on water side	0.002 hr-ft²F/BTU		$\left(\frac{0.35}{m^2 \cdot K/kW}\right)$
Velocity in exchanger tubes	5-8 fps		(1.5-2.5 m/s)

Provision shall be made for complete venting and draining.

• **2.1.9** Control of the sound level of all equipment furnished shall be a joint effort of the purchaser and the vendor. Unless otherwise specified, the equipment furnished by the vendor shall conform to the requirements of API Standard 615 and the maximum allowable sound level specified by the purchaser.

• **2.1.10** All electrical components and installations shall be suitable for the area classification and grouping specified by the purchaser on the data sheets, the requirements of NFPA Bulletin Number 70, Articles 500 and 501, and specified local codes.

• **2.1.11** The purchaser shall specify and furnish to the vendor with his inquiry any local rules or regulations that will apply. All equipment furnished shall comply with all federal, state, and local codes and regulations applicable at the installation site.

2.2 Casings

2.2.1 All pressure parts shall be at least suitable for operation at the most severe conditions of coincident pressure and temperature expected with the specified steam conditions.

2.2.2 The hoop-stress values used in the design of the casing shall not exceed the values given for maximum allowable stress values in tension, as specified in Section VIII, Division I of the ASME Code, at maximum operating temperature of the material used.

2.2.3 Turbine casing shall be split axially. Multistage turbine casing may also be split radially between high- and low-pressure portions.

2.2.4 Axial split casings shall use a metal-to-metal joint (with suitable joint compound) tightly maintained by suitable bolting. Gaskets (including string type) shall not be used on the axial joint. Gasketed joints when used between the end covers and the cylinder of radial split casings shall be securely maintained by confining the gaskets.

2.2.5 Casing and supports shall be designed to have sufficient strength and rigidity to limit change of alignment due to pressure, torque, or allowable piping forces and moments to 0.002 inches (50 micrometers) at the coupling flange. (See 2.5.) Turbines shall have centerline supports to maintain proper alignment with connected equipment. Supports and alignment screws shall be rigid enough to permit the machine to be moved using the lateral and axial jackscrews.

2.2.6 Jackscrews, guide rods, and casing alignment dowels shall be provided to facilitate disassembly and reassembly. When jackscrews are used as a means of parting contacting faces, the mating face shall be relieved (counterbored or recessed) to prevent a leaking joint or improper fit caused by marring. Lifting lugs or eyebolts shall be provided for lifting only the top half of the casing. Methods of lifting the assembled machine shall be specified by the vendor. Guide rods shall be of sufficient length to prevent damage by casing or casing studs during assembly.

2.2.7 Steam chest and casing shall be provided with drains for complete drainability.

2.2.8 Tapped holes in pressure parts shall be held to a minimum. Sufficient metal, in addition to metal allowance for corrosion, shall be left around and below the bottom of drilled and tapped holes in pressure sections of casings to prevent leakage.

2.2.9 Bolting shall be furnished as follows:

1. Details of threading shall conform to ANSI B1.1.
2. Studs are preferred to cap screws.
3. Preferably, blind stud holes should not be drilled deeper than necessary to allow a tap depth of 1½ times the major diameter of the stud. Studs for use in blind stud holes shall have the first 1½ threads removed from each end.
4. Clearance shall be provided at bolting locations to permit the use of socket- or box-type wrenches.
5. ANSI B31.3, Paragraph 309, shall govern the material limits for pressure bolting based upon the actual bolting

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temperature. Nuts shall conform to ASTM A 194, Grade 2H (or ASTM A 307, Grade B, casehardened, where space is limited).
6. Socket-, slotted-nut, or spanner-type bolting shall not be used unless specifically approved by purchaser.

2.3 Casing Appurtenances

2.3.1 NOZZLES AND DIAPHRAGMS

2.3.1.1 All nozzle rings shall be replaceable. Welded-in design is acceptable with prior purchaser approval.

2.3.1.2 All other stationary blading shall be mounted in replaceable diaphragms or segments.

• **2.3.2** When specified a sentinel warning valve shall be supplied on the turbine casing. It shall be set at 5 pounds per square inch gage (0.35 bar) on condensing turbines. For noncondensing turbines it shall be set at either 10 pounds per square inch gage (0.7 bar) or 10 percent above maximum exhaust pressure, whichever is greater.

2.4 Casing Connections

2.4.1 Inlet and outlet connections shall be flanged, or machined and studded, oriented as specified in the data sheets, and shall be suitable for steam conditions as specified and defined in 1.4, unless otherwise agreed upon by the purchaser.

2.4.2 All casing openings for pipe connections shall be not less than ¾-inch NPS and shall be flanged or machined and studded. Where flanged or machined and studded openings are impractical, threaded or socket-weld openings are permissible in sizes ¾ inch through 1½ inch (NPS) inclusive. Tapped openings and bosses for pipe threads shall conform to ANSI B16.5. Pipe threads shall be taper threads conforming to ANSI B2.1. Threaded or socket-weld openings shall be installed as follows:

1. A pipe nipple, preferably not over 6 inches (152 millimeters) long, shall be screwed or welded into the opening.
2. Pipe nipples shall be a minimum of Schedule 160 seamless for sizes 1 inch and smaller and Schedule 80 for 1½ inch.
3. The pipe nipple shall be provided with a welding-neck or socket-weld flange.
4. The threaded connection shall be seal-welded. However, seal welding is not required on cast-iron equipment, for instrument connections, or where disassembly is required for maintenance. Seal-welded threaded connections shall be made up without thread lubricants, tape, or compound. The seal weld shall consist of at least two passes, with the throat of the finished weld ⅜ inch (9.5 millimeters) maximum and shall cover all exposed threads. No undercutting shall be permitted.

2.4.3 Openings for pipe sizes of 1¼, 2½, 3½, 5, 7, and 9 inches shall not be used.

2.4.4 Tapped openings not connected to piping shall be plugged with solid steel, long shank plugs furnished in accordance with ANSI B16.11. Plugs that may later require removal shall be of corrosion-resistant material. Threads shall be lubricated. Tape shall not be applied to threads of plugs inserted into oil passages. Plastic plugs are not permitted.

2.4.5 Flanges up to and including 24 inches shall conform to ANSI B16.1 or B16.5 as applicable, except as follows:

1. Flanges shall be full- or spot-faced on the back. Cast-iron flanges shall be flat-faced and shall be Class 250 thickness for 8 inches and smaller.
2. Flat-faced flanges are acceptable on all exhaust connections. Purchaser approval is required for other application of flat-faced flanges on steel cases.
3. Flanges which are thicker or have a larger outside diameter than required by ANSI are acceptable.
4. Flanges larger than 24 inches shall conform to API Standard 605 (1967) or to MSS SP44.
5. Integral flanges shall be machined such that the eccentricity of the bolt circle with respect to the bore shall not exceed ⅛ inch.
6. If flanges other than the above are offered, details shall be approved by the purchaser and all required mating parts shall be supplied by the vendor.
7. On fabricated casings, flanges may be either ring or hub type, provided they conform with the above standards.

2.4.6 Machined and studded connections shall conform to ANSI B16.1 or B16.5 for facing and drilling requirements. Studs and nuts shall be furnished installed. Connections larger in pipe diameter than covered by ANSI shall meet the requirements of 2.4.5, item 4.

2.4.7 All purchaser's connections shall be accessible for disassembly without moving the machine.

2.4.8 Pipe-flange gaskets shall be spiral-wound metal, asbestos-filled, or metal-jacketed for steam temperature above 500F (260C) or steam pressure above 400 pounds per square inch gage (28 bar). The manufacturer's standard gasket can be used below these limits.

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2.5 External Forces and Moments

Turbines shall be designed to withstand external forces and moments at least equal to the values calculated in accordance with NEMA SM-23. In some cases, these allowable forces and moments can be increased after considering such factors as location and degree of turbine supports, nozzle length and degree of reinforcement, and casing configuration and thickness.

2.6 Rotating Elements

2.6.1 ROTORS

2.6.1.1 Rotors (other than integrally forged) shall be assembled so that movement of the disk relative to the shaft is prevented when operating at any speed permitted by protective devices. Wheels shall be keyed to the shaft and assembled with a shrink fit.

2.6.1.2 Rotors shall be designed to operate safely at momentary speeds up to 110 percent of the trip setting at operating temperature.

2.6.1.3 Purchaser approval is required for built-up rotors when blade tip velocities exceed 825 feet per second (250 meters per second) at maximum continuous speed or when stage inlet steam temperatures exceed 825F (441C).

2.6.1.4 All rotors shall be clearly marked with a unique identification number.

2.6.2 SHAFTS

2.6.2.1 Shafts shall be accurately finished throughout their entire length and shall be ground at coupling, bearing areas, and sealing areas for carbon-ring packing. The finish of these ground surfaces shall not exceed 16 microinches (0.4 micrometers) root mean square (rms).

2.6.2.2 The surface area to be observed by the vibration probe shall be concentric with the bearing journals and free from stencil and scribe marks or any other surface discontinuity such as an oil hole or keyway. This area shall not be metallized or plated. The final finish shall be 16 to 32 microinches rms (0.4 to 0.8 micrometers), preferably obtained by honing or burnishing (not by grinding). This area shall be properly demagnetized or otherwise treated so that the combined total of mechanical and electrical runout shall not exceed 25 percent of the maximum allowed peak-to-peak vibration amplitude or 0.25 mils (6 micrometers), whichever is greater.

2.6.2.3 Keyways shall have fillet radii, conforming to ANSI B17.1.

2.6.2.4 Shafts shall be protected by corrosion-resistant material under carbon-ring packing for casing end glands; the manufacture's application method shall be stated on the data sheet.

2.6.2.5 Flanged shaft ends for coupling fits shall conform to API Standard 671 for dimensional tolerances.

2.6.3 BLADING

2.6.3.1 Excitation of in-phase tangentials, out-of-phase tangentials, axial, torsional, and any other high-response modes of each blade row by multiples of up to 15 times running speed, by nozzle passing frequency, and by twice nozzle passing frequency shall be verified by Campbell or similar diagrams as outside the specified operating speed range. The verifying data shall be provided to the user. If this is not feasible, blade-stress levels developed at any specified driven equipment operation shall be low enough to insure trouble-free operation if resonant vibration occurs within the operating range. This shall be verified by Goodman or equal diagrams, copies of which shall be provided to the user. Blades shall be designed to withstand operation at resonant frequencies during normal warm up.

2.6.3.2 All blades shall be mechanically suitable for operation (including for transient conditions) over the specified speed range. The vendor will assume that torque varies as speed squared unless otherwise notified by the purchaser.

2.7 Shaft Seals

● **2.7.1** Outer glands shall be sealed with replaceable labyrinth packing unless carbon-ring packing is specified by the purchaser.

2.7.2 When carbon-ring packing is specified, it shall be used only when the rubbing speed at the shaft-sealing surface is less than 160 feet per second (49 meters per second). The number of carbon rings shall be determined by the service and venting requirements, with 25 pounds per square inch (1.72 bar) being the maximum average differential pressure per active sealing ring. Springs for carbon packing shall be inconel (heat-treated after cold coiling) or equal material. Consideration shall be given to operating steam-temperature variations in establishing packing clearances.

2.7.3 Sealing of interstage diaphragms shall be by replaceable labyrinth packing.

2.7.4 Glands operating at less than atmospheric pressure shall be designed for admission of steam to seal against air leakage. Piping with relief valve, pressure gages, regula-

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tors, and other necessary valves shall be provided to interconnect the end glands. Piping shall have one common connection to the purchaser's sealing steam supply. The admission of sealing steam shall be automatically controlled throughout the load range. The normal operating sealing steam supply shall preferably come from a positive pressure section of the turbine.

2.7.5 A separate vacuum device shall be furnished, when specified. The device shall be connected to the glands to reduce external leakage and possible contamination of the bearing oil. The device shall be mounted and connected when specified. (See 3.5.) Appendix A shows a typical gland vacuum system.

2.7.6 All piping and components of shaft seal and vacuum systems shall be sized for 300 percent of the calculated new clearance leakage.

2.8 Dynamics

2.8.1 CRITICAL SPEED

2.8.1.1 Critical speeds correspond to resonant frequencies of the rotor-bearing support system. The basic identification of critical speeds is made from the natural frequencies of the system and of the forcing phenomena. If the frequency of any harmonic component of a periodic forcing phenomenon is equal to, or approximates, the frequency of any mode of rotor vibration, a condition of resonance may exist; if resonance exists at a finite speed, that speed is called a critical speed.

Concern of this specification is with the actual critical speeds rather than various calculated values. Actual critical speeds are not calculated undamped values but are critical speeds confirmed by test-stand data in addition to calculated damped values above test speeds or by rotor excitations.

2.8.1.2 The turbine vendor shall make a lateral critical speed analysis and determine that the critical speeds of the turbine are compatible with the operating speed range. The vendor shall include his assumptions regarding bearing and support stiffness used for calculation of the lateral criticals in the summary report and, when specified, shall submit calculated response curves of deflection versus revolutions per minute for varying amounts of unbalance.

A forcing phenomenon or exciting frequency may be less than, equal to, or greater than the synchronous frequency of the rotor. Such forcing frequencies may include, but not be limited to—

1. Unbalance in the rotor system.
2. Oil film frequencies.
3. Internal rub frequencies.
4. Blade, vane, nozzle, diffuser passing frequencies.
5. Gear-meshing and side-band frequencies.
6. Coupling misalignment frequencies.
7. Loose rotor system component frequencies.
8. Hysteresis and friction whirl frequencies.
9. Boundary layer (vortex shedding) frequencies.
10. Acoustical or aerodynamical frequencies.
11. Start-up condition frequencies, for example, speed detents under inertial impedances or torsional deflections contributing to torsional resonances.

2.8.1.3 Bearing housing resonance shall not occur within the specified operating speed range.

2.8.1.4 Actual critical speeds shall not encroach upon specified operating speed ranges. Amplification factor *(AF)* shall not exceed 8 while going through criticals. Values of *AF* less than 5 are preferred. This measurement shall be recorded on deceleration (coast down) with the slow-roll (300-600 revolutions per minute), total runout (electrical and mechanical) subtracted by vectorial runout compensation. This recorded shaft vibration data shall include speed, peak-to-peak displacement, and phase.

2.8.1.5 The separation margin (see Figure 1) of encroachment from all lateral (includes stiff-shaft and bending) modes shall be—

1. Twenty percent over the maximum continuous speed for stiff-shaft rotor systems.
2. Fifteen percent below any operating speed and 20 percent above the maximum continuous speed for flexible shaft rotor systems.

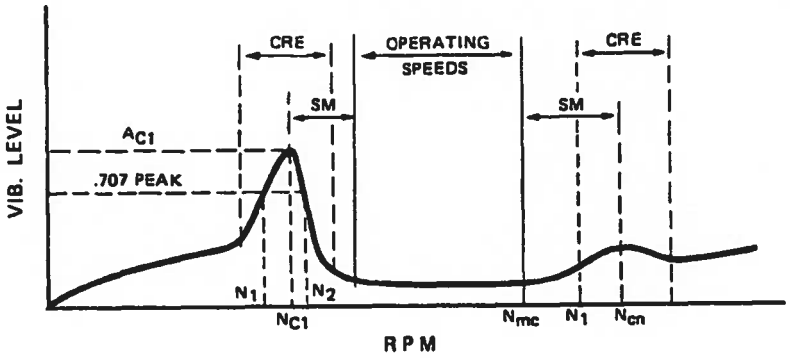
2.8.1.6 The specified separation margin in 2.8.1.5 is intended to prevent the overlapping of the resonance response envelope into the operating speed range. The margin of separation shall be increased to prevent any actual overlap incurred.

2.8.1.7 Slow-roll, start-up, and shutdown of rotating equipment shall not cause any damage as critical speeds are passed.

2.8.1.8 If the critical speed as calculated or revealed during mechanical testing falls within the specified operating speed range or fails to meet the separation margin requirements after practical design efforts are exhausted, the unit vendor shall demonstrate an insensitive rotor design. This insensitivity must be proved by operation on the test stand at the critical speed in question with the rotor unbalanced. Vibration shall not exceed twice the API vibration limits (see 2.8.2.4). During the sensitivity tests, the vibration amplitude increment shall be based on the difference between the major axes of the orbits formed by synchronous X-Y signals in the balanced and unbalanced runs. Deflec-

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N_{c1} = Rotor 1st critical, center frequency, cycles per minute
N_{cn} = Critical speed, nth
N_t = Trip speed
N_{mc} = Maximum continuous speed, 105 percent
N_1 = Initial (lesser) speed at .707 x peak amplitude (critical)
N_2 = Final (greater) speed at .707 x peak amplitude (critical)
N_2-N_1 = Peak width at the "half-power" point

AF = Amplification Factor $= \frac{N_{c1}}{N_2 - N_1}$
SM = Separation Margin
CRE = Critical Response Envelope
A_{c1} = Amplitude @ N_{c1}
A_{cn} = Amplitude @ N_{cn}

NOTE: Curve shape is for illustration only and does not necessarily represent any actual rotor response plot.

Figure 1 — Rotor Response Plot

tions of the rotor shall not exceed design clearances. After proof of rotor's insensitivity, the unbalanced weights shall be removed. The amount of unbalance shall be—

$$U_B \text{ in ounce-inch} = 5\left[\begin{matrix}\text{API residual}\\ \text{unbalance}\\ \text{limit}\end{matrix}\right] = 5\left(\frac{56.347\ W_r}{N_{mc}^2}\right)$$

Where:

W_r = rotor weight, pounds
N_{mc} = maximum continuous speed, 105 percent

Modal analysis shall be used in the placement of the unbalanced weights as mutually agreed upon by the equipment builder and purchaser.

1. Translational modes shall be unbalanced by placement of unbalanced U_B totally at the center of the rotor.
2. Conical modes shall be unbalanced by placing ½ U_B near each bearing but 180 degrees out-of-phase.

CAUTION—Excitation by unbalancing at the coupling shall not exceed ½ U_B.

2.8.2 VIBRATION AND BALANCE

2.8.2.1 Major parts of the rotating element, such as impellers and balancing drums, shall be dynamically balanced.

2.8.2.2 The rotating element shall be multiplane dynamically balanced during assembly. This shall be accomplished after the addition of no more than two major elements to a flexible rotor with normal speed above 6000 revolutions per minute. Rotors with normal speeds of 6000 revolutions per minute and below shall have a minimum of three balancing planes. Balance correction shall only be applied to new elements.

Rotors with single keys for couplings shall be balanced with half keys in place. The maximum allowable unbalanced force at any journal at maximum continuous speed shall not exceed 10 percent of the static loading of that journal.

• **2.8.2.3** When specified, after completion of the final balancing of the assembled rotating element, the sensitivity of the rotor-balancing machine system shall be reported in the following manner:

1. The vendor shall supply to the purchaser the journal static loading data in pounds.
2. The maximum allowable residual unbalance shall be calculated from the following equation:

$$\text{ounce} - \text{inch} = \frac{56{,}347 \times (\text{Journal Static Loading})}{rpm^2}$$

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Where:

rpm = maximum continuous speed in revolutions per minute.

3. The ounce – inch value calculated shall be divided by the radius from the shaft center to the approximate center of the balance correction area in the correction plane nearest that journal. The result is the weight of a test weight (W) to be applied at that radius in that plane.
4. The rotating element shall be spun in the balancing machine, and the resultant machine readout shall be recorded.
5. Further test weights of $W/4$, $2W$, and $4W$ shall be prepared and sequentially applied at the exact same point. The resultant readout shall also be recorded.
6. A smooth curve shall be plotted of the test weights versus the balancing machine readouts. This curve shall be continued smoothly to a theoretical zero readout (see Figure 2).
7. The process shall be repeated for the other end of the rotating element, and a similar curve shall be plotted.
8. If either of the two extrapolated curves intersects the ordinate at $W/4$ or higher (see Figure 2, Curve B), it shall be concluded that the sensitivity of the balancing machine system for use with the rotating element is either marginal or unacceptable, and the rotating element shall then be rebalanced using a more sensitive machine.

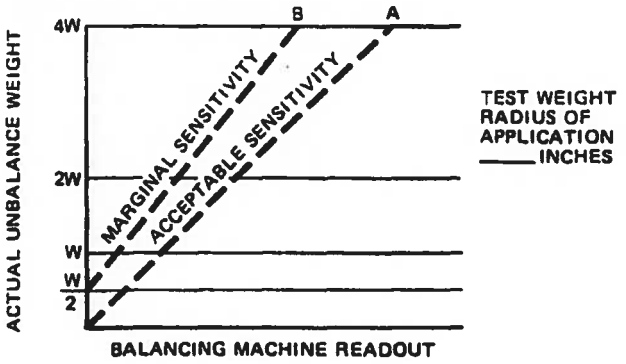


Figure 2 — Balancing Machine Response Check

2.8.2.4 During the shop test of the assembled machine operating at maximum continuous speed or at any other speed within the specified operating speed range, the double amplitude of vibration in any plane measured on the shaft adjacent and relative to each radial bearing shall not exceed the following value or 2.0 mils (50 micrometers), whichever is less:

$$\text{Unfiltered double amplitude including runout}^{9} \text{ (mils)} = \text{Vibration} + \text{Runout} = \sqrt{\frac{12{,}000}{rpm}} + 0.25\sqrt{\frac{12{,}000}{rpm}}$$

Where:

rpm = maximum continuous speed in revolutions per minute.

At the trip speed of the driver, the vibration shall not exceed the above value plus 0.5 mil (13 micrometers).

Electrical runout can be deduced by slow-rolling the rotor in bearings or vee-blocks while measuring runout with a proximity probe and a dial indicator at the same shaft location.

2.9 Bearings and Bearing Housing

2.9.1 RADIAL BEARINGS

Hydrodynamic radial bearings shall be split for ease of assembly, of precision-bored, sleeve or pad type, with steel-backed babbitted replaceable liners, pads, or shells. The bearing design shall suppress hydrodynamic instabilities and provide sufficient damping to limit rotor vibration to maximum specified amplitudes (see 2.9.2.4) while operating at any support critical frequency. The liners, pads, or shells shall be in horizontally split housings and shall be replaceable; removal of the top half of the casing of a horizontally split machine or the head of a radially split unit shall not be required for replacement of these elements.

2.9.2 THRUST BEARINGS

2.9.2.1 Hydrodynamic thrust bearings shall be steel-backed, babbitted, multiple-segment type, designed for equal-thrust capacity in both directions, and arranged for continuous pressurized lubrication to each side. Both sides shall be tilting-pad type incorporating a self-leveling feature assuring that each segment carries an equal share of the thrust load regardless of segment thickness.

The thrust bearing shall be adjustable for axial positioning of the rotor and shall be located on the inlet end of the turbine.

2.9.2.2 Oil outlets from thrust-bearing cartridges shall be tangential.

2.9.2.3 Thrust collars may be integral, or separate and replaceable. Replaceable collars shall be positively locked

[9] Shaft runout is the total indicator reading in a radial direction when the shaft is rotated in its bearings. If the vendor can demonstrate that "electrical runout" due to shaft material anomalies is present, the combined total mechanical and electrical runout shall not exceed 25 percent of the specified test level or 0.25 mil (6 micrometers) minimum, whichever is greater.

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to the shaft to prevent fretting. Integral collars shall be provided with a minimum of ⅛-inch additional stock (3 millimeters) on each thrust face for remachining should the collar be damaged. Both faces of the collar shall have a surface finish not exceeding 16 microinches (0.4 micrometers). The axial total indicated runout of either face shall not exceed 0.0005 inches (13 micrometers).

2.9.2.4 Thrust bearings shall be sized for continuous operation under thrust loads from the turbine-steam thrust plus margin for fouling, step thrust, and the thrust transmitted by the coupling as specified in 2.9.2.4, items 1 and 2.

Thrust bearings shall be selected at no more than 50 percent of the bearing's ultimate load capability (based on test data) for the specific application regarding speed, lubricant, and lubricant temperature (see Appendix B for bearing selections based on loading, speed, and thrust-shoe area). The sizing basis shall be reviewed with and approved by the purchaser.

1. For gear-type couplings, the external force shall be calculated from the following formula:

$$\text{Coupling thrust in pounds} = \left(\frac{0.25 \times 63{,}000}{rpm \times D}\right) \times hp$$

Where:

hp = rated horsepower
rpm = rated speed in revolutions per minute
D = shaft diameter at the coupling in inches

2. Thrust loads for diaphragm-type couplings shall be calculated on the basis of the maximum allowable deflection permitted by the coupling manufacturer.

2.9.3 BEARING HOUSINGS

2.9.3.1 Bearing housings and shells shall be axially split.

2.9.3.2 Bearing housings shall be equipped with labyrinth-type end seals and deflectors where the shaft passes through the housing; lip-type seals shall not be used. Deflectors shall be made of nonsparking materials. The labyrinth deflector design shall effectively retain oil in the housing and prevent entry of foreign material into the housing.

2.9.3.3 Bearing housing for hydrodynamic bearings designed for pressure lubrication shall be arranged to minimize foaming. The drain system shall be adequate to maintain the oil and foam level below shaft-end seals. Oil temperature rise through the bearing and housings shall not exceed 50F or 28C under the most adverse specified operating conditions when inlet oil temperature is 110F (43C). Where oil inlet temperatures exceed 120F (49C), special consideration shall be given to bearing design, oil flows, oil composition, and allowable temperature rise.

2.9.3.4 Access shall be provided for shaft vibration measurements to be taken with a hand-held probe adjacent to each journal bearing.

2.9.3.5 Bearing housings for machines with rated speeds above 5000 revolutions per minute and fitted with sleeve-type bearings shall preferably be designed to accept tilting-pad radial bearings without additional machining. Vendor will advise in proposal whether or not such interchangeability is provided.

2.9.3.6 Provision shall be made for mounting two radial vibration sensor probes in each bearing housing and two axial position probes at the thrust end of each machine. Probe locations and mounting details shall be as specified in API Standard 670.

2.9.3.7 Access shall be provided for a hand-held tachometer to verify the turbine speed.

2.9.4 GROUNDING

All condensing turbines shall be provided with a minimum of two grounding brushes on the shaft. Brushes shall be located for convenient replacement and in an area not flooded by oil. The brushes shall be a standard size and grade suitable for the service. The vendor shall include with his proposal drawings showing the number and location of the brushes.

2.10 Lubrication and Control Oil System

2.10.1 When specified, the turbine vendor shall furnish the lube or control oil system, or both.

2.10.2 If the lube and control oil system is furnished by others, the turbine vendor shall—

1. Furnish the level of steady and transient oil flow-pressure required, the degree of filtration required, and the maximum heat load imposed.
2. Furnish piping to single feed connections for each pressure level. One drain connection shall be provided for all oil to be returned to the reservoir.

2.10.3 All oil systems and components shall conform to API Standard 614.

• **2.10.4** Where oil is supplied from a common system to two or more machines (such as compressor, gear, and motor), the oil characteristics shall be specified on the data sheet by the purchaser on the basis of mutual agreement with all vendors supplying equipment served by the common oil system. The usual lubricant used in a common oil system is a hydrocarbon oil with approximate viscosity of 150 Saybolt Universal Seconds (SUS) at 100

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F (32 square millimeters per second at 38C) and 43 SUS at 210F (5.3 square millimeters per second at 99C).

2.10.5 Any points requiring grease lubrication shall have suitable extension lines for accessibility.

2.11 Materials

2.11.1 GENERAL

2.11.1.1 Materials of construction shall be manufacturer's standard for the specified operating conditions, except as required by the data sheets or this specification. See 3.6 for auxiliary piping material requirements.

2.11.1.2 Materials shall be identified in the proposal with the applicable ASTM, AISI, ASME, or SAE numbers. When no such designation is available, the manufacturer's name and the physical properties and chemical composition of the material shall be presented for the purchaser's approval.

2.11.1.3 The vendor shall specify ASTM optional tests and inspection procedures necessary to ensure that materials are satisfactory for the service. Such tests and inspections shall be listed in the proposal. Purchaser should consider specifying additional tests and inspections, especially for critical service (see 4.2.3).

2.11.2 TURBINE PARTS

2.11.2.1 Pressure parts shall be of steel if the maximum steam conditions to which they may be subjected exceed 250 pounds per square inch gage (17 bar) or 500F (260C) total temperature or if the normal steam temperature exceeds 450F (232C), except that exhaust casings of noncondensing turbines shall be of steel if the maximum exhaust pressure exceeds 75 pounds per square inch gage (5.2 bar) or no-load exhaust temperature exceeds 500F (260C). Suitable alloy steel shall be used where the maximum steam temperatures exceed 750F (400C).

2.11.2.2 Material for turbine parts shall be forged steel for wheels and shaft and 11-13 percent chromium, titanium, or monel for the following parts: nozzles, closing pieces, blading (rotating and stationary), shrouding, and steam strainers.

2.11.2.3 External parts subject to rotary or sliding motions (such as control linkage joints, valve stems, springs and adjusting mechanism) shall be of corrosion-resistant materials suitable for site environment.

2.11.2.4 Minor parts not identified (nuts, springs, washers, gaskets, keys, and so forth) shall have equal corrosion resistance to specified parts in the same environment.

2.11.2.5 All plate material used in the manufacture of pressure parts shall be one of those listed under Subsection C, ASME Code, Section VIII, Division 1, and shall comply with the general requirements of Subsection A and the specific requirements of Subsections B and C as pertaining to the method of fabrication and material.

• **2.11.2.6** For ambient temperatures below −20F (−29C), when specified, carbon steels shall have an impact strength at the lowest temperature sufficient to qualify under the minimum Charpy-V notch impact energy requirements of the ASME Code, Section VIII, Division 1, UG-84.

2.11.2.7 It shall be the responsibility of the purchaser to fully specify the presence of unusual enviromental conditions or the contamination of the process steam with agents which might promote stress corrosion cracking. The vendor shall recognize, however, that the average steam system includes contaminants such as sodium hydroxide, chlorides, sulfates, phosphates, copper, and lead and shall consider this in selecting materials.

• **2.11.2.8** When specified, the vendor shall list in the proposal all steam path components (valves, seat blades, shrouds, closing pieces and pins, damping wires, wheels, bolting, and so forth) that will be above Rockwell C-22 in hardness together with the hardness range of such components.

2.11.3 CASTINGS

2.11.3.1 Castings shall be sound and free of shrink holes, blow holes, cracks, scale, blisters, or other similar injurious defects. Surfaces of castings shall be cleaned by sandblasting, shotblasting, pickling, or any other standard method. All mold-parting fins and remains of gates and risers shall be chipped, filed, or ground flush.

2.11.3.2 The use of chaplets in pressure castings shall be held to a minimum. They shall be clean and corrosion-free (plating permitted) and of a composition compatible with the casting.

2.11.3.3 Ferrous castings shall not be repaired by welding, peening, plugging, burning-in, or impregnating, except as follows:

1. Weldable grades of steel castings may be repaired by welding, using a qualified welding procedure based on the requirements of Sections VIII or IX of the ASME Code.
2. Cast gray iron or nodular iron may be repaired by plugging within the limits specified in ASTM A 278 or A 395, respectively. The drilled holes for a plug shall be carefully examined by dye penetrant to ensure the removal of all defective material. All necessary repairs not covered by ASTM shall be subject to approval by the purchaser.

2.11.3.4 Fully enclosed cored voids, including voids closed by plugging but excluding cooling jackets, are prohibited.

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2.11.4 WELDING

2.11.4.1 All construction and repair weldings shall be performed by welders qualified in accordance with Section IX of the ASME Code. All welding procedures shall conform to Section VIII of the ASME Code.

2.11.4.2 Magnetic particle inspection shall be performed on all repair welds in accordance with 2.11.5.3.

2.11.4.3 Pressure-containing castings under internal pressure greater than 300 pounds per square inch gage (20 bar) made of wrought materials or combinations of wrought and cast materials shall have all accessible weld sufaces inspected by magnetic particle or liquid penetrant examination after back chipping or gouging and again after stress relieving.

2.11.4.4 All pressure-containing welds, including case-to-horizontal and vertical-joint flanges, shall be full penetration.

2.11.4.5 All fabricated casings regardless of thickness shall be post-weld heat-treated.

2.11.5 INSPECTION

• **2.11.5.1** When specified, castings exposed to internal pressure shall be examined by radiography in accordance with the requirements given in the *ASME Boiler and Pressure Vessel Code*, Section VIII, Division 1, Appendix VII. Radiography would normally be considered for steam conditions exceeding 1250 pounds per square inch gage (86 bar) or 900F (482C). With purchaser's approval, ultrasonics may be used in lieu of radiography.

• **2.11.5.2** When specified, weldments exposed to internal pressure shall be examined by radiography in accordance with the requirements given in the ASME Code, Section VIII, Division 1, Subsection B, Paragraph UW-51.

2.11.5.3 Magnetic particle inspection shall be in accordance with Appendix VI of the ASME Code, Section VIII, Division 1. When magnetic particle inspection as described in ASTM E 109 (dry) or ASTM E 138 (wet) is required, acceptability of defects shall be based on a comparison with the photographs associated with ASTM E 125. For each type of defect, the degree of severity shall not exceed the limits in the following list:

Type	Degree
I	1
II	2
III	2
IV	1
V	1
VI	1

• Regardless of these generalized limits, it shall be the vendor's responsibility to review the design limits of all castings in the event more stringent requirements are specified. Where defects exceed the limits imposed above, they shall be cleaned out to meet the quality standards referenced above as determined by additional magnetic particle inspection, prior to repair welding.

2.11.5.4 Dye-penetrant inspection shall be in accordance with Appendix VIII of the ASME Code, Section VIII, Division 1.

2.11.5.5 Forgings used for turbine shafts, discs, and rotors with integrally forged discs shall be inspected by ultrasonic methods per ASTM A 418.

2.11.5.6 When specified or performed, a heat stability check shall be in accordance with ASTM A 472.

2.11.5.7 Records of all heat treatment and radiographs, fully identified, whether performed in the normal course of manufacture or as part of a repair procedure, shall be kept available for 5 years for review by the purchaser.

2.12 Nameplates

2.12.1 Nameplates and rotation arrows shall be 18-8 stainless steel or monel, securely fastened by pins of similar material, and located for easy visibility.

2.12.2 The following data, as a minimum, shall be clearly stamped on the nameplate. The contract or data sheets shall specify English, metric units, or both.

Vendor's name

Serial number

Size and type

Rated power and speed

First critical speed [11]

Second critical speed [11, 12]

Maximum continuous speed

Overspeed trip [10]

Normal and maximum inlet steam temperature and pressure

Normal and maximum exhaust steam pressure

Purchaser's equipment item number [13]

[10] Overspeed trip setting; see 4.3.4.1 on testing.

[11] If critical speed values are not obtained by test, the word "calc" shall be stamped beside the number. Any critical speeds below maximum continuous speed shall be determined on the test stand.

[12] May be omitted for stiff-shaft turbines.

[13] May be on separate nameplate if there is insufficient space on rating nameplate.

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SECTION 3—ACCESSORIES

3.1 Couplings and Guards

3.1.1 COUPLINGS

• **3.1.1.1** A flexible coupling between driver and driven equipment shall be supplied by the manufacturer of the driven equipment unless otherwise specified on the data sheets. When the turbine vendor supplies items that are contiguous, he shall furnish a flexible coupling located between the items.

• **3.1.1.2** Couplings shall conform to API Standard 671. The coupling make, type, and mounting arrangement shall be agreed upon by the driver and driven equipment vendors, and the purchaser. A spacer coupling shall be used unless otherwise specified.

3.1.1.3 Information on shaft, keyway dimensions (if any), and shaft end movements due to end play and thermal effects shall be furnished to the vendor supplying the coupling.

3.1.1.4 The horsepower rating of the coupling to shaft juncture shall be at least equal to the horsepower rating of the coupling.

3.1.1.5 The coupling purchaser shall supply an idling adaptor as required under 4.3.4.3, item 6.

3.1.2 GUARDS

3.1.2.1 A guard conforming to API Standard 671, Appendix A shall be placed over all exposed couplings furnished by the turbine vendor.

3.1.2.2 External feed and drain lines for lubricated couplings shall be classed as auxiliary piping.

3.1.3 MOUNTING

3.1.3.1 Coupling mounting shall conform to API Standard 671.

3.1.3.2 The turbine half of the coupling shall be mounted by the turbine manufacturer. See 4.3.4.3, item 6 for running tests.

3.2 Mounting Plates

3.2.1 The turbine shall be provided with soleplates or a baseplate, as specified on the data sheets.

3.2.2 In the following paragraphs the term mounting plate refers to both baseplates and soleplates:

1. The equipment feet and the mounting plates shall have machined surfaces.
2. The equipment feet and mounting plates shall be equipped with vertical jackscrews and shall be drilled with pilot holes for use in final doweling. Mounting plates for installation on concrete shall be supplied with leveling screws.
3. Where the equipment supported weighs more than 1000 pounds (453 kilograms), the mounting plates shall be furnished with horizontal jackscrews, the same size or larger than the vertical jackscrews.
4. Stainless steel laminated shim packs, ⅛-inch (3-millimeter) minimum, shall be furnished between the equipment feet and the mounting plates by the mounting plate vendor.
5. Anchor bolts will be furnished by the purchaser. They shall not be used to fasten machinery to the mounting plates.
6. The hardware for attaching turbine vendor-furnished components to the mounting plates and for leveling shall be supplied by the mounting plate vendor.
• 7. When epoxy grout is indicated on the data sheet, the vendor shall precoat all of the grouting surfaces of the mounting plates with epoxy resin applied to degreased white metal.

3.2.3 BASEPLATES

• Where a baseplate is specified, the data sheet must show the extent. Unless otherwise specified, the baseplate shall be a single, fabricated steel unit.

• **3.2.3.1** If specified by the purchaser, the baseplate shall be suitable for column mounting (that is, of sufficient rigidity to be supported at specified points) without continuous grouting under structural members.

3.2.3.2 The baseplate shall be provided with lift lugs for a four-point lift. Lifting the baseplate complete with all equipment mounted shall not permanently distort or otherwise damage the baseplate or the machinery mounted on it.

3.2.3.3 The bottom of the baseplate between structural members shall be open. Accessibility for grouting and filling with concrete shall be provided. The mounting pads on the bottom of the baseplate shall be in one plane to permit use of a single-level foundation with continuous grouting under the baseplate. Use of sub-soleplates shall be mutually agreed to between purchaser and vendor. If required, sub-soleplates (leveling plates) shall be provided by the vendor.

• **3.2.3.4** When specified on the data sheet, decking shall be provided on the top of the baseplate covering all walk and work areas.

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3.2.3.5 When the purchaser approves oil reservoirs located in the baseplate, the reservoir shall conform to the requirements of API Standard 614.

3.2.4 SOLEPLATES

Where soleplates are specified, they shall be of adequate size and thickness to transmit the loads from the equipment feet to the foundation and to provide adequate working clearance for anchor bolts and jackscrews.

3.3 Gear Units

Gear units shall conform to API Standard 613.

3.4 Controls, Protective Devices, and Instrumentation

3.4.1 GENERAL

• **3.4.1.1** Instrumentation and installation shall conform to any detailed specifications in the purchaser's inquiry or order, or both. When no detailed specifications are furnished, instrumentation and installation shall conform to the requirements of API Standard 614, API Standard 670, and API RP 550.

• **3.4.1.2** Unless otherwise specified, all controls and instrumentation shall be suitable for outdoor installation.

3.4.2 CONTROLS

• **3.4.2.1** The governor shall be mechanical/hydraulic or electrical/hydraulic, as specified.

3.4.2.2 Electrical/hydraulic governors shall have no mechanical connection to the turbine rotor. The speed-sensing system shall be redundant. Failure of one speed pickup shall trigger an alarm. Details of the electrical/hydraulic governor shall be mutually agreed upon between the purchaser and vendor. Unless otherwise specified, the purchaser will furnish the electrical power supply.

3.4.2.3 Unless otherwise specified, the speed governor shall control the turbine at all speeds within the specified range. Other control modes (such as those for extraction/induction machines) shall be mutually agreed upon between the vendor and purchaser.

3.4.2.4 Governor systems shall be designed to prevent the unit from exceeding the turbine trip speed with an instantaneous loss of rated load.

3.4.2.5 The governor shall be furnished with a device which allows manual override of the maximum governor speed setting for overspeed trip testing.

3.4.2.6 Unless otherwise specified, the speed governor for mechanical drive application shall conform to NEMA SM-23, Class D, with an adjustable speed range of 30 percent below maximum continuous speed. Speed adjustment shall be provided at the governor by a hand speed
• changer. When a control signal is specified, the governor also shall be furnished with an actuator arranged to receive a signal from an external control device for setting the control point of the governor. Control signal range shall be 3 to 15 pounds per square inch for pneumatic instruments and 4 to 20 milliamperes for electronic instruments. The actuator shall be arranged so that—

1. The speed of the turbine shall vary linearly with the control signal.
2. A signal change of plus or minus ½ percent of the signal range shall cause a change in the turbine speed.
3. The required speed range shall correspond to the full range of the control signal specified.
• 4. An increase in the control signal shall increase the turbine speed, unless otherwise specified on the data sheet.
5. The control-signal actuation, or failure of the signal or actuator, shall prevent neither the governor from limiting the speed to the maximum permissible nor manual adjustment with the hand speed changer.

• **3.4.2.7** Unless otherwise specified, the speed governor for generator drive application shall conform to NEMA SM-23, Class C, with an adjustable droop. Speed adjustment shall be provided at the governor by a hand speed changer. Remote speed (load) adjustment actuators shall be provided when and as specified. Details of generator drive controls shall be mutually agreed upon between the vendor and purchaser.

3.4.2.8 Unless otherwise specified on the data sheets, a minimum of two hand-operated control valves shall be provided on all single inlet valve turbines. These valves shall be used to meet the requirements of 2.1.2, item 2
• and for economical operation at less than the rated load of the turbine. When specified, hand valves shall be automatically operated.

3.4.3 PROTECTIVE DEVICES

• **3.4.3.1** A separate, mechanically operated trip and throttle valve shall be furnished. The valve and trip mechanism shall be rugged and positive acting, and shall be designed so that the valve can be operated manually to regulate steam admission.

It shall close by spring action and stop the turbine either upon overspeeding, lube-oil pressure falling below a safe level, manual tripping, or when actuated by hydraulic, pneumatic, solenoid, or other required or specified protective devices. When a solenoid device is specified, it

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shall close the trip and throttle valve when energized unless otherwise specified. The trip valve shall be capable of being reset with maximum inlet steam pressure on the line. Seating surfaces and stem shall be of corrosion-resistant material. On turbines with an exhaust pressure less than atmospheric, consideration shall be given to having an exhaust vacuum breaker actuated by the hydraulic or electrical trip systems. Details of such a system shall be mutually agreed upon by the purchaser and vendor.

3.4.3.2 The trip and throttle valve shall be equipped with a steam strainer designed to prevent in-service failure. The strainer shall be removable without dismantling any of the inlet steam piping.

3.4.3.3 Unless otherwise specified, actuation of a trip device shall cause the steam inlet valves to close in addition to closing the trip and throttle valve. Facilities needed to permit one operator to restart the turbine shall be provided. Vendor shall describe the provisions in the proposal.

3.4.3.4 Trip and throttle valves shall be capable of being partially stroked (either manually or hydraulically) to exercise the valve without interrupting the operation of the turbine.

3.4.3.5 When specified, the turbine shall be provided with an electrically operated, field adjustable, overspeed trip independent of the mechanical overspeed trip.

3.4.3.6 On extraction or extraction-admission turbines, the vendor shall supply an additional protective device on all possible steam supply lines. Make, model, and location of the device(s) shall be mutually agreed upon by purchaser and vendor.

3.4.4 INSTRUMENT PANEL

3.4.4.1 When specified, a panel shall be provided and shall include all panel-mounted instruments for the driven unit and driver. Such panels shall be designed and fabricated in accordance with purchaser's description. The instruments on the panel shall be clearly visible to the operator from the driver control point. A lamp test push button shall be provided. Instruments mounted on the panel will be specified by the purchaser and may include those from Table 1.

TABLE 1 — Instrument Gages

1. Inlet steam pressure gage
2. Inlet and exhaust steam temperature gage
3. Exhaust steam pressure gage
4. Steam-ring pressure gage on single-valve turbines (oil filled with inlet snubber)
5. First-stage steam pressure gage on multistage turbines (oil filled with inlet snubber)
6. Extraction pressure gage (oil filled with inlet snubber)
7. Pressure gage for first stage after extraction section (oil filled with inlet snubber)
8. Seal steam pressure gage
9. Lube-oil pressure gage
10. Control (governor) oil pressure gage
11. Speed indicator
12. Gages for driven equipment
13. Alarms and indicator lights
14. Bearing temperature indicator
15. Vibration and axial position readout equipment
16. Other instruments specified for panel mounting

3.4.5 OTHER INSTRUMENTATION

- **3.4.5.1** Unless otherwise specified, two digital readout speed indicators shall be furnished. Readout shall be to nearest 10 revolutions per minute. One indicator shall be mounted in the local panel and one shall be supplied to the purchaser for remote mounting.

- **3.4.5.2** Switches, control devices, and annunciator display units shall be furnished and mounted by the vendor, as specified by the purchaser, and may include those from Table 2.

TABLE 2 — Alarms and Shutdowns

	Alarm Only	Anticipatory Alarm With Annunciated Shutdown	Annunciated Shutdown
Radial Shaft Vibration	X		
Axial Position Thrust	X		
Speed			X
Bearing Temperature	X		
Unit Shutdown	X		
Lube-Oil Pressure		X	
Lube-Oil Temperature	X		
High Turbine Extraction Pressure	X		
High Exhaust Pressure	X		

- **3.4.5.3** When and as specified, iron-constantan thermocouples or resistance temperature detectors (RTDs) shall be furnished and installed in two "active side" thrust-bearing shoes approximately 180 degrees apart, two "inactive side" thrust-bearing shoes approximately 180 degrees apart, and two in each radial bearing, oriented adjacent to each other at approximately load line, for connection to purchaser's temperature monitoring equipment. The thermocouple or RTD leads shall be adequately protected (stainless-steel armor shielding or equivalent) against abrasion, chafing, and vibration within the bearing housing, shall exit from housing through an oil-tight fitting, and shall terminate in a thermocouple head mounted on the outside of the bearing housing.

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3.4.5.4 Each alarm and shutdown switch shall be furnished in a separate housing. "Single-pole, double-throw" switches shall be used.

• **3.4.5.5** "Open" (deenergize) to alarm and "close" (energize) to trip electric switches shall be furnished unless otherwise specified.

• **3.4.5.6** The vendor shall furnish a "first-out" type of annunciator when an annunciator system is specified.

3.4.5.7 Alarm and trip-switch settings shall not be adjustable from outside the housing. Pressure elements shall be of 18-8 stainless steel. Alarm and shutdown switches shall be arranged to permit testing of the control circuit, including the actuating element where possible, without interfering with normal operation of the equipment. The vendor shall provide a clearly visible light on the panel to indicate when trip circuits are in the test bypass mode. The vendor shall furnish with the proposal complete descriptions of alarm and shutdown testing facilities to be provided, that is, high-pressure alarms should be equipped with valved test connections so that a portable test pump can be used to raise the pressure. In addition, shutdown systems should be provided with electrical disconnect switches or other suitable means to permit testing without shutdown of the unit.

3.4.5.8 All instruments and controls other than shutdown-sensing devices shall be installed with sufficient valving to permit removing instruments and controls while the system is in operation.

3.4.6 ELECTRICAL SYSTEMS

• **3.4.6.1** Electrical power supply characteristics for motors, heaters, and instrumentation will be specified by the purchaser. Pilot lights shall be provided on the incoming side of each supply circuit to indicate that circuits are energized. Lights shall be installed on the control panels.

3.4.6.2 All electrical equipment located on the unit or on any separate panel shall be suitable for the hazard classification specified. Electrical starting and supervisory controls may be either ac or dc.

3.4.6.3 All power and control wiring within the confines of the baseplate shall be resistant to heat, moisture, and abrasion. Stranded conductors shall be used within the confines of the baseplate and other areas subject to vibration. Measurement and remote-control panel wiring may be solid conductor. Where rubber insulation is used, a neoprene or high-temperature thermoplastic sheath shall be provided for insulation protection. All wiring shall be suitable for environmental temperatures.

3.4.6.4 To facilitate maintenance, liberal clearances shall be provided for all energized parts (such as terminal blocks and relays) on all equipment. Clearances required for 600-volt service also shall be provided for all lower voltages.

• **3.4.6.5** All electrical materials shall be corrosion-resistant and nonhygroscopic insofar as possible. When specified for tropical location, all material shall be given the following treatments:

1. All insulated impregnated parts (such as coils and windings) shall be impregnated with antifungicidal varnish.
2. Surfaces to be painted shall have at least one coat of corrosion-resistant paint.
3. Unpainted surfaces shall be protected against corrosion by plating or other suitable coating.

3.4.6.6 Where practicable, design and location of all thermocouples shall permit replacement while the unit is operating. Thermocouple lead wires shall be installed as continuous leads between thermowell and control panel. Conduit runs from thermocouple heads to pull boxes located on the baseplate shall be provided.

3.4.6.7 All wiring, including thermocouple leads, within limits of the baseplate shall be installed in rigid metallic conduits, properly bracketed to minimize vibration. Conduits may terminate (and in the case of thermocouple heads shall terminate) with a flexible metallic conduit of sufficient length to permit access to the unit for maintenance without removal of the conduit. If thermocouple heads will be exposed to temperatures above 140F (60C), a ¾-inch bronze hose having four-wall interlocking construction and asbestos joints with packed-on (heat-proof) couplings shall be used.

3.4.6.8 For Division 2 locations, flexible metallic conduits shall have a liquid-tight thermosetting or thermoplastic outer jacket and approved fittings. For Division 1 locations, approved flexible connectors shall be provided.

3.4.7 VIBRATION AND POSITION DETECTORS

3.4.7.1 Unless specified otherwise, noncontacting vibration and axial-position transducers shall be supplied, installed, and calibrated in accordance with API Standard 670.

3.4.7.2 Unless specified otherwise, vibration and axial-position monitors shall be supplied and calibrated in accordance with API Standard 670.

• **3.4.7.3** The data sheet shall indicate when noncontacting probes are not required or if velocity or acceleration transducers are required.

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3.5 Gland Vacuum System

3.5.1 The gland condenser shall be sized based on 2.7.6. It shall have a steel shell, ⅝-inch/18 BWG Admiralty tubes, fixed tube sheets, water on the tube side. U-tubes are not acceptable. Water side conditions shall conform to 2.1.8. Design and test pressures shall be as follows:

	Design	Hydrotest
Shell	Full Vacuum	20 psig (1.4 bar)
Tubeside	150 psig (10.3 bar)	225 psig (15.5 bar)

3.5.2 The steam ejector shall have a steel body and an AISI 416 stainless steel replaceable steam nozzle. It shall be sized based on 2.7.6.

3.6 Piping and Appurtenances

3.6.1 Piping systems are defined as those auxiliary systems attendant to the turbine and auxiliaries which carry the following:

Lubricating oil
Control oil
Steam including gland sealing
Cooling water
Drained or vented fluids
Air or nitrogen

3.6.2 All piping and appurtenances as defined by 3.6.1 shall conform to API Standard 614 with the following exceptions and modifications:

1. Flexible sections, not more than 18 inches (46 centimeters) long, consisting of Type 321 stainless steel, annular corrugated, close pitch; stainless-steel braided hose may be used in oil drains. These flexible sections shall be welded (not brazed) to the end fittings. For maritime environments, the flexible sections shall be covered with a sealed, heavy duty, reinforced synthetic casing not subject to attack by chlorides.
2. All components in the piping system that contain steam shall be fabricated of steel. They shall be a minimum of ¾-inch NPS. Threaded connections, except for instruments, shall be held to a minimum and shall be seal-welded.
3. Slip-on flanges are permitted in all piping except oil piping.
4. The requirements for gaskets given in 2.4.8 also apply to piping.
5. Where space does not permit the use of pipe, ½-, ¾-, and 1-inch seamless steel tubing in accordance with ASTM A 192 or stainless steel tubing in accordance with ASTM A 269 may be furnished with steel fittings. Minimum wall thickness shall be 0.065 inch (1.65 millimeters) for ½ inch, 0.095 inch (2.41 millimeters) for ¾ inch, and 0.109 inch (2.76 millimeters) for 1 inch. Make and model of fittings shall be subject to approval by purchaser.

3.7 Insulation and Jacketing

• **3.7.1** When specified, turbines shall be insulated and jacketed. The insulation and jacketing shall extend over all portions of the turbine casing that may reach a normal operating temperature of 165F (74C) or higher.

3.7.2 The insulation shall maintain a jacket surface temperature of not more than 165F (74C) under normal operating conditions. Jackets and insulation shall be designed to minimize possible damage during removal and replacement.

3.8 Weather Protection

• **3.8.1** When specified, weather protection shall be furnished. The purchaser shall specify whether the installation is indoors (heated or unheated) or outdoors (with or without a roof). The purchaser also shall specify the weather conditions in which the turbine unit must operate, including maximum and minimum temperatures, precipitation (if outdoors), unusual dust problems, and other related factors.

• **3.8.2** For outdoor installations, either in the open or under a roof, turbine units and auxiliaries shall be winterized for the particular refinery atmosphere, including temperature, as specified by the purchaser. The vendor shall list in the proposal any special protection required for the purchaser's guidance.

3.9 Special Tools

3.9.1 When tools and fixtures not commercially available are required in order to disassemble, assemble, or maintain the turbine unit, the tools and fixtures shall be included in the quotation and be furnished as part of the initial supply of the turbine.

3.9.2 When special tools are provided, they shall be packaged in separate, rugged boxes and marked as "special tools for (tag/item number)." Each tool shall be tagged, indicating its intended use.

3.10 Turning Gear

• **3.10.1** Turning gear shall be provided whenever specified by the purchaser or required by the vendor.

NOTE: Turning gear normally is not required but should be considered for large turbines particularly when a requirement to restart quickly within 8 hours of shutdown is expected.

3.10.2 The turning device shall be designed so that it cannot be engaged unless lube-oil pressure is established and so that it must be manually engaged to operate the turning gear and automatically disengaged when the turbine is placed into service.

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3.10.3 The turning gear may be driven by a steam turbine, electrical motor, hydraulic motor, or pneumatic motor. The turning-gear driver shall be in accordance with the appropriate API standard.

SECTION 4 — INSPECTION AND TESTING

4.1 General

4.1.1 The vendor shall provide the purchaser with advance notification of certain shop inspections and testing, as outlined in the purchase order or other agreements. The purchaser's representative shall have entry after prior notification by the purchaser to all vendor and subvendor plants where work upon or testing of the equipment is in progress. It shall be the responsibility of the vendor to notify subvendors of the purchaser's inspection requirements.

4.1.2 In each instance the actual number of calendar days considered to be sufficient advance notice shall be established by the mutual consent of the purchaser and the vendor.

4.1.3 The purchaser will specify the extent of his participation in the testing.

1. *Witness* means that a hold shall be applied to the production schedule and the test carried out with the purchaser in attendance. It usually implies a double test. Because of the hold and the repetitive test, a charge is made for a witness test.
2. *Observed* means that the purchaser requires notification of the test timing. However, the test is performed as scheduled, and if the purchaser is not present the vendor can proceed to the next step. Since only one test is scheduled, the purchaser should expect to be in the factory longer than for a witness test. Normally, no additional charge is made for an observed test.

4.2 Inspection

4.2.1 The vendor shall keep the following data available for examination by the purchaser or his representative upon request:

1. All necessary certification of material, such as mill test reports.
2. Purchase specifications for all items on bills of material.
3. Running test data to verify that the requirements of the specification are being met.
4. Results of quality assurance tests.

4.2.2 All painting of surfaces of pressure-containing parts shall be deferred until the specified inspection of that part is completed.

4.2.3 The purchaser shall specify—

1. Parts which shall be subject to surface and subsurface inspection.
2. Type of inspection required, such as magnetic particle, dye penetrant, radiographic, ultrasonic, and so forth.

4.2.4 During assembly of the system and prior to testing each component (including cast-in passages of these components), all piping and appurtenances shall be cleaned by pickling or by another appropriate method to remove foreign materials, corrosion products, and mill scale.

4.2.5 The oil system furnished shall meet the cleanliness required by API Standard 614.

4.2.6 When specified, the purchaser may make an inspection for cleanliness of the equipment and all piping and appurtenances furnished by or through the vendor prior to the installation of nozzle blocks and steam-chest covers, the closure of openings in vessels or exchangers, or the assembly of piping.

4.2.7 When specified, hardness of parts, welds, and heat-affected zones shall be verified as being within the allowable values by testing of the parts. Method, extent, documentation, and witnessing of test shall be mutually agreed to by purchaser and vendor.

4.2.8 When requested, purchaser's representative shall have access to the vendor's quality control program for review.

4.2.9 Acceptance of shop tests shall not constitute a waiver of requirements to meet the field tests under specified operating conditions, nor does inspection relieve the vendor of his responsibilities in any way.

4.3 Testing

4.3.1 GENERAL

4.3.1.1 All turbines shall be tested in accordance with 4.3.3 and 4.3.4. Other tests which may be specified by the purchaser are described in 4.3.5.

4.3.1.2 Equipment for specified tests shall be provided by the vendor.

4.3.1.3 The vendor shall notify the purchaser not less than 5 days prior to the date the equipment will be ready for test.

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• 4.3.2 WITNESS TESTS

The purchaser shall specify tests for which a witness will be required.

4.3.3 HYDROSTATIC TESTS

4.3.3.1 Pressure-containing parts (including auxiliaries) shall be tested hydrostatically with liquid at a minimum of 1½ times the maximum allowable working pressure but not less than 20 pounds per square inch gage (1.38 bar).

4.3.3.2 Where the steam temperature exceeds 650F (344C), the values of the test pressure shall be multiplied by a factor obtained by dividing the maximum allowable hoop-stress value of the material at room temperature by that of the corresponding stress value at steam temperature. For hoop-stress values, refer to Section VIII, Division 1 of the ASME Code. The data sheet shall list the actual hydrostatic test pressures.

4.3.3.3 Tests shall be maintained for a sufficient period of time to permit complete examination of parts under pressure. The hydrostatic test shall be considered satisfactory when no leaks are observed for a minimum of 30 minutes. Large and heavy castings may require a longer testing period to be agreed upon by purchaser and vendor. Seepage past internal test closures required for segmental case testing and operation of test pump to maintain pressure will be accepted.

4.3.3.4 Use of a sealant compound on the casing joints is acceptable during hydrotest.

4.3.4 MECHANICAL RUNNING TEST

4.3.4.1 Overspeed trip device(s) shall be checked and adjusted until three consecutive nontrending trip values within plus or minus 1 percent of nominal trip setting are obtained.

4.3.4.2 The mechanical running test of the turbine shall be conducted in the following sequence:

1. Operate at speeds from zero to maximum continuous speed in approximately 10 percent increments and run at maximum continuous speed until bearings and lube-oil temperatures have stabilized. Caution should be exercised when operating at or near critical speeds.
2. Increase the speed to approximately 1 percent below the trip speed and run for a minimum of 15 minutes.
3. Reduce speed to maximum continuous and run for 4 hours.

4.3.4.3 The following basic requirements shall be met for the running test:

1. The contract shaft seals and bearings shall be used in the machine for the mechanical running test.
2. All oil pressures, viscosities, and temperatures shall be at the same operating values recommended in the manufacturer's operating instructions for the specific unit under test. Lube-oil flow rates for bearing housings shall be determined. (Accepted methods other than flow meter may be used.) Test-stand oil filtration shall be 10 micrometers or better.
3. The casing and oil system shall be free from leakage during the test.
4. All warning, protective, and control devices shall be checked and adjustments made as required.
5. Along with the vendor's standard test log of vibration amplitudes, the following test recordings shall be made at each shaft radial probe:
   a. While the turbine is operating at maximum continuous speed, recordings shall be made of vibration amplitude versus frequency. As a minimum, these sweeps shall cover a range of 0.25 to 8 times maximum continuous speed with a maximum of 90,000 cycles per minute (1500 Hertz). If the amplitude of any discreet nonsynchronous vibration exceeds 20 percent of the allowable vibration as defined in 2.8.2.4, the vendor and purchaser shall mutually agree on the requirements for additional testing, if any, and suitability for shipment.
   b. Plots showing vibration amplitude and phase angle versus speed for acceleration and deceleration shall be made before and after the 4-hour run. The speed range covered by these plots shall be from zero to the specified trip speed.
6. The test shall be made with half-coupling(s) and idling adapter(s) in place (resulting in a moment equivalent to that of the contract half-coupling plus one-half the coupling spacer).
7. Steam conditions shall be as close to specified normal as practicable.

4.3.4.4 During the running test, the mechanical operation of all equipment being tested and test instrumentation shall be satisfactory. Unfiltered vibration measurements shall be recorded throughout the operating speed range and shall not exceed the limits of 2.8.2.4.

4.3.4.5 During the running test, the location(s) of any lateral critical speed(s) from zero to maximum continuous speed shall be verified.

4.3.4.6 If the critical speed(s) encroach upon the specified operating speed range (see 2.8.1.5), rotor insensitivity must be demonstrated in accordance with 2.8.1.8.

4.3.4.7 In addition to the vendor's standard instrumentation, the following are required during the mechanical test:

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1. All purchased vibration probes and oscillator demodulators shall be in use during the mechanical running test. If vibration probes are not furnished by the compressor vendor or the purchased probes are not compatible with shop readout facilities, then shop probes and readout meeting accuracy requirements of API Standard 670 shall be used.
2. Shop test facilities shall include instrumentation with the capability of continuously monitoring and plotting revolutions per minute, peak-to-peak displacement, and phase angle (X-Y-Y'). Presentation of vibration displacement and phase marker also shall be by oscilloscope.

• **4.3.4.8** When specified, FM-tape recordings of data shall be made. Details of what data are to be recorded shall be mutually agreed upon by purchaser and vendor.

4.3.4.9 Vibration characteristics determined by using instrumentation specified in 4.3.4.7 shall serve as the basis for machine acceptance or rejection (2.8.2.4).

4.3.4.10 Control systems shall be demonstrated to the extent practicable.

1. Speed governor and any speed-regulating devices shall be tested for smooth performance over the operating speed range, no-load stability, and response to control signal.
2. Data shall be taken including, but not limited to, adjustable governor-response speed range, sensitivity, and linearity of relationship between speed and control signal.

4.3.4.11 All bearings shall be removed, inspected, and reassembled after completion of the mechanical running test.

4.3.4.12 If replacement or modification of bearings or seals or dismantling of the case to replace or modify other parts is required for improvement of mechanical or performance deficiencies, the initial test will not be acceptable and the final shop tests shall be run after such replacements or corrections are made.

4.3.4.13 When spare rotors are ordered to permit concurrent manufacture, each spare rotor also shall be given a mechanical run test.

4.3.5 OPTIONAL TESTS

4.3.5.1 Optional tests shall be developed jointly by the purchaser and the vendor.

• **4.3.5.2** When specified, performance tests shall be conducted in accordance with ASME PTC-6. If this is not practicable, the vendor shall state the conditions under which the vendor proposes to conduct the tests. Details shall be subject to negotiation.

• **4.3.5.3** The purchaser shall specify in the inquiry or in the order if any of the following shop tests shall be performed.

1. *Complete unit test*: Such components as compressors, gears, drivers, and auxiliaries that make up a complete unit shall be tested together during the mechanical running test. A separate auxiliary test may be performed with the purchaser's approval.
2. *Sound level test*: The sound level test shall be performed in accordance with API Standard 615.
3. *Mechanical test*: Mechanical test with lube-oil temperature varied from high to low limits allowed by vendor's operating manual. This may be done during the 4-hour standard mechanical test run.
4. *Auxiliary equipment test*: Auxiliary equipment such as oil systems and control systems shall be tested in the vendor's shop.

4.3.6 TEST DATA RECORDS

4.3.6.1 The vendor shall make the following test data available for examination by the purchaser or his representative upon request. All data shall be suitably identified by rotor and unit identification numbers.

1. Certified copies of hydrostatic and running test records, including any optional tests purchased.
2. Recorded or logged (at 15-minute intervals) test data including:
   a. Data acquired during mechanical run:

Time	Displacement and phase angle
Speed	Overspeed trip speed
Oil pressures (in and out)	Governor linearity and adjustability
Oil temperatures (in and out)	Cooling water temperatures
Steam conditions	Bearing metal temperatures (if detectors specified)
Vibration measurements	Oil flows
Vibration frequency sweep	
Location of critical speed(s)	

   b. Data acquired during performance run as agreed to under 4.3.5.2.
   c. Data acquired during other tests as agreed.
3. Balance and runout data.

4.4 Preparation for Shipment

4.4.1 GENERAL

• **4.4.1.1** Equipment shall be suitably prepared for the type and mode of shipment specified including blocking of the rotor when necessary. The preparation shall be suitable for a period of 6 months of outdoor storage from the time of shipment, such that disassembly is not required (except for bearing inspection) prior to operation.

• **4.4.1.2** If storage is contemplated for longer periods, the purchaser shall consult with the vendor regarding recommended procedures to be followed.

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4.4.1.3 Preparation for shipment shall be made after all testing and inspection of the equipment has been accomplished, and the equipment has been released for shipment by the purchaser. The preparation shall include the following as a minimum:

1. All exterior surfaces with the exception of machined surfaces shall be given a coat of the manufacturer's standard paint.
2. All exterior machined surfaces shall be coated with a suitable rust preventative.
3. The interior of the equipment shall be sprayed or flushed with a suitable rust preventative which is removable by solvent. This shall be done through all openings while the machine is slow-rolled.
4. After having been thoroughly cleaned, all internal areas of bearings and carbon-steel oil systems' auxiliary equipment such as reservoirs, vessels, and piping shall be coated with a suitable oil-soluble rust preventative.
5. All flanged openings shall be provided with metal closures of $\frac{3}{16}$-inch (5-millimeter) minimum thickness with rubber gaskets. At least four full-diameter bolts shall be used for flanged openings. For studded openings, all studs and nuts needed for the intended service shall be used for securing closures. See 2.4.6.
6. All threaded openings shall be provided with steel caps or solid shank steel plugs. In no case shall nonmetallic plugs or caps (such as plastic) be used.
7. All turbines supplied without self-supporting baseplates shall be bolted to a shipping skid formed of heavy timbers suitable for sling-lift or forklift truck handling. Larger turbines shall have supports as required by mode of transportation and handling. Lifting points and lifting lugs shall be clearly indicated.
8. Equipment shall be identified with item and serial numbers. All material shipped separately shall be identified with securely affixed corrosion-resistant and weather-resistant metal tags indicating the item and serial
• number for which it is intended. Turbines with carbon seal rings shall have the seal rings removed by the vendor and shipped in a separate container. A red tag on the turbine shall read as follows:

CARBON SEALS MUST BE INSERTED BEFORE THIS TURBINE IS STARTED.

9. All components (individual pieces as well as packaged sets) shipped with mounted preassembled piping, tubing, or wiring shall comply with Occupational Safety and Health Administration (OSHA) requirements and shall carry outside securely affixed large, red, all-weather tags stating in bold letters:

THIS SYSTEM HAS BEEN PREASSEMBLED AND TESTED FOR OPERABILITY AND SAFETY, AND COMPLIES WITH ALL REQUIREMENTS OF OSHA (OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION), AND SHALL NOT BE DISTURBED BY UNAUTHORIZED PERSONNEL.

• 10. When a spare rotor is purchased, the rotor shall be suitably prepared for unheated, indoor, vertical or horizontal storage, as specified (or as recommended by the vendor), for a period of at least 3 years. The rotor shall be treated with a rust preventative and shall be housed in a vapor-barrier envelope with slow-release vapor-phase inhibitor. The rotor shall be suitably crated for domestic or export shipment, as specified. Suitable $\frac{1}{8}$-inch (3-millimeter) minimum thickness lead sheeting shall be used between the rotor and the cradle at the support areas. The rotor shall not be supported at the journals.

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SECTION 6 — VENDOR'S DATA

6.1 Proposals

6.1.1 The vendor's proposal shall include the information listed in items 1 through 22. (See Appendix C for a sample data sheet.)

1. Copies of purchaser's data sheet(s) with complete vendor's information entered.
2. Complete performance curves to fully define the envelope of operations and the specific guarantee point. For extraction turbines, an extraction diagram showing the throttle flow at all combinations of horsepower and extraction rate and based on operation at normal speed shall be supplied.
3. Utility requirements (such as steam, water, electricity, air, gas, and lube oil) including the quantity of lubricating oil required at supply pressure, the heat load to be removed by the oil, driver ratings, and net loads. (Approximate data shall be defined clearly as such. If possible, this information shall be entered on the data sheets.)
4. Net and maximum operating weights, maximum erection weight with identification of the item. (This data shall be stated individually where separate shipments, packages, or assemblies are involved. If possible, this data shall be entered on the data sheets.)
5. Each proposal shall be accompanied by a preliminary dimensional outline drawing which shows location of inlet, extraction, and exhaust openings.
6. Separate, typical cross-sectional drawings that show details of construction, including governor, inlet and extraction valves, design of blades and shroud attachments, shaft sealing details, bearings, internal construction, and nonreturn valves. Descriptive literature and data, modified to provide a clear understanding of the proposed equipment, shall be provided.
7. A diagram with a description of the governing system.
8. Schematic diagrams of the lube- and control-oil systems and the gland-sealing system when furnished by turbine vendor.
9. A specific statement that the system and all components are in strict accordance with purchaser's specifications. (If not in strict accordance with the specifications, the vendor shall include a specific list detailing and explaining every deviation. Deviations may include alternative designs or systems equivalent to and guaranteed for the specified duties.) (See 1.2.1.)
10. The maximum allowable casing pressures and temperatures shall be entered on the data sheet(s).
- 11. If specified, proposal shall indicate approximate potential maximum horsepower output of the unit at normal steam conditions that could be obtained by field modification. The required field modifications shall be described in general, such as: valve, nozzle, diaphragm, or blade changes with no changes to the rotor or casing.
12. An explicit statement of any deviations from the specified guarantee and warranty (see Section 5).
13. A statement of the fixed number of weeks to effect shipment after receipt of the order and all engineering data. Separate time shall be stated for manifold shipment as in the case of separate package, assemblies, or multiple units.
14. A statement of vendor's promised time(s), after order placement, for transmittal of the contract data (see 6.2). (This information shall be presented in the form of an explicit schedule. See Appendix D.)
15. Vendor's proposal shall include separate prices for spare rotors, bearings, seals, labyrinths, and couplings (when furnished). The vendor shall state specific price incentives for ordering these spares within a stated period to permit concurrent manufacture.
16. An itemized list of the special tools included in the offering.
17. The test and procedures for materials as required by 2.11.1.3.
18. A statement of the rate for furnishing a competent erection supervisor, plus an estimate of the length of time the supervisor's services might be required under normal conditions.
19. *Optional tests*: Separate prices for optional tests shall be included when these are specified.
20. A description of special requirements as outlined in 2.6.2.4, 2.9.3.5, 2.9.4, 2.11.1.2, 2.11.2.8, 3.4.3.3, 3.4.5.7.
21. An outline of all necessary special weather and winterizing protection required by turbine and the auxiliaries for startup, operation, and idleness. The vendor shall quote separately the protective items the vendor proposes to furnish.
22. A statement of the length of time the information outlined in 4.2.1 will be retained.

6.2 Contract Data

6.2.1 GENERAL

6.2.1.1 The following paragraphs specify the information to be furnished by the vendor. This data shall be tabulated on the vendor data requirement sheet (see Appendix D) and forwarded to the address(es) as noted on the order. The data sheet shall detail the schedule for transmission of data as agreed to at time of order and the number and type of copies of all data required.

6.2.1.2 The data shall be identified on transmittal (cover) letters and in title blocks/pages with (a)

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purchaser/user corporate name; (b) job project number; (c) equipment name and tag number; (d) specific purchase order number; (e) any other identification specified in the purchase order; and (f) vendor's identifying shop order number, serial number, or whatever reference required to completely identify return correspondence.

NOTE: For driven equipment, the applicable API or purchaser specification shall apply to data required.

6.2.2 DRAWINGS

6.2.2.1 The purchaser shall state in the inquiry and in the order whether or not manufacturing is to proceed without purchaser's review of the drawings.

6.2.2.2 Review of the vendor's drawings shall be made promptly after receipt by the purchaser. However, such review shall not constitute permission to deviate from any requirements in the order unless specifically agreed upon in writing. After the drawings have been reviewed, the vendor shall furnish certified copies in the quantity specified. All drawings and reproducibles supplied shall be suitable for microfilming.

6.2.2.3 The drawing(s) furnished shall contain sufficient information that when combined with the manuals covered in 6.2.5, the purchaser may properly install, operate, and maintain the ordered equipment. The following details are minimum:

1. Overall dimension(s) and weight(s) for each separately installed piece. Include maintenance clearance and weight-handling capability required for erection and maintenance.
2. Maximum allowable forces and moment loadings for all major flanged connections.
3. Complete information for adequate foundation design by purchaser. Include grouting details, foundation bolt sizes and locations, weight distributions for each bolt/sub-soleplate location, and any unbalanced forces or moments generated by the unit(s) in the operating range specified and the location of the center of gravity.
4. Schematic diagrams and bills of material of all auxiliary systems within the vendor's entire scope of supply (oil piping, controls, and so forth). The diagrams shall include schematic drawings of control systems, shaft sealing, lubricating-oil systems, and dimensional outline drawings for accessories and instrument panels. The drawings shall include and identify all components by make, type, size, capacity, pressure rating, materials, and other data as applicable.
5. The direction of rotation.
6. The size, type, location, and identification of all purchaser's connections, including vents, drains, lubricating oil, instruments, and controls. Vendor's plugged connections shall be identified.
7. When applicable, the make, size, and type of couplings. The location and coverage of coupling guards shall be indicated.
8. Special weather protection and climatization features supplied by vendor and required by purchaser, if any.
9. Auxiliary or equipment furnished by vendor for mounting by purchaser.
10. Thermal and mechanical movements of casing and shafts.
11. Rigging provisions for removal of the top half of casing weighing more than 300 pounds (136 kilograms).

Each drawing shall have a title block in the lower right-hand corner with certification, reference to all identification data specified in 6.2.1, revision number, date, and drawing title, all visible when drawing is folded to 8½ x 11 inches.

6.2.2.4 *Drawing list*: Complete drawing list of all drawings shall be included with first-issue major drawings. This list will contain titles and schedule for transmission of all drawings vendor will furnish.

6.2.3 CURVES

6.2.3.1 Vendor shall provide complete performance curves to encompass the map of operations, with limitations indicated, if any. The curves shall include:

1. Steam flow versus horsepower at normal and rated speed and steam conditions.
2. Steam flow versus first-stage pressure for multistage machines or nozzle-bowl pressure for single valve, single-stage machines.
- 3. Exhaust steam temperature (or enthalpy if specified) versus horsepower with normal inlet steam conditions and exhaust pressure for all units except condensing.
4. Speed versus torque.
5. First-stage pressure versus thrust.

Vendor shall supply steam-flow correction curves for the following variations from normal conditions:

1. Speed (over governor speed range).
2. Maximum and minimum specified exhaust pressure.
3. Maximum and minimum specified inlet pressure.
4. Maximum and minimum specified initial temperature.

- The purchaser shall specify his requirements for additional curves or maps for extraction/induction turbines. The following are examples:

1. Extraction/induction flow versus power and throttle flow for extraction/induction turbines.
2. Exhaust enthalpy versus extraction/induction flow.
3. Steam flow versus first stage after extraction section pressure for extraction turbines.

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6.2.4 DATA

6.2.4.1 Vendor shall provide a completed as-built data sheet.

6.2.4.2 The vendor shall make available to the purchaser the following information:

1. Vendor's physical and chemical data from mill reports (or certification) of steel pressure parts and forgings.
2. Certified shop logs of mechanical run test.
3. A record of shop test data maintained for at least 18 months from date of shipment. When specified, certified copies of test data shall be submitted to the purchaser prior to shipment.
4. The required number of copies of the lateral critical analysis summary report (no later than 3 months after order date) which shall include but not be limited to the following:
   a. Complete description of the method used in determining the critical speed.
   b. Graphic display of critical speeds versus operating speeds.
   c. Graphic display of support stiffness and its effect on critical speeds.
   d. Graphic display of rotor response to unbalanced weight.
5. Campbell and Goodman diagrams or their equivalent as required by 2.6.3.1.
6. Any additional data required shall be specified on the contract information data sheet along with a schedule, as agreed at time of order.

6.2.5 INSTALLATIONS AND INSTRUCTION MANUALS

6.2.5.1 Manufacturers shall provide sufficient written instructions and a cross-referenced list of all drawings to enable purchaser to correctly install, operate, and maintain the complete equipment ordered. This information shall be compiled in a manual (or manuals) with a cover sheet containing all reference identifying data required in 6.2.1.2 and an index sheet containing section titles and a complete list of referenced and enclosed drawings by title and drawing number.

6.2.5.2 Any special information required for proper installation design and actual installation that is not on the drawings shall be compiled in a manual that is separate from operating and maintenance instructions. This manual should be forwarded at a time as mutually agreed in the order, but not later than final-issue prints. It should contain information such as special alignment or grouting procedures, utility specifications (including quantity), and all installation design data.

6.2.5.3 The manual containing operating and maintenance data shall be forwarded not later than 2 weeks after successful completion of all specified tests. The manual shall include a section, if required, to cover special instructions for operations at specified extreme environmental conditions, such as temperature. The following shall be included in this manual:

1. Instructions covering startup, normal shutdown, emergency shutdown, operating limits, and routine operational procedures.
2. Description of turbine construction features and the functioning of component parts or systems such as governing, lubrication, overspeed protection, and steam seal leak-off.
3. Outline drawings, sectional drawings, schematic sketches, and illustrative sketches in sufficient detail to identify all parts and to show clearly the operation of all equipment and components and the methods of inspection and repair. Standardized sectional drawings are acceptable only if they represent the actual construction of the turbine.
4. Instructions for erecting, piping, aligning (including expected thermally induced shaft centerline movement between 60F (16C) ambient temperature position and that at normal operating steam temperature and pressure), and preparing the turbines and auxiliary equipment for use.
5. Description of rigging procedures, methods of disassembly, repair, adjustment, inspection, and reassembly of turbines and auxiliaries.
6. The following maintenance information:
   a. Maximum and minimum labyrinth, packing, and bearing clearance.
   b. Instruction for measuring and adjusting cold clearances.
   c. Rotor-float allowance and axial clearance tolerances of the blades.
   d. Interference fit on thrust collars, sleeves, couplings, and wheels.
   e. Runout and concentricity tolerances on shafts, wheels, blading, and thrust collars.
   f. Balancing tolerances.
7. The following reassembly information:
   a. Bolting sequence and torque values for such items as casing bolting, thrust-collar lock nuts (when used), and internal bolting.
   b. Reassembly sequences together with required inspection checks.
   c. Adjustment procedures to achieve required positions, clearances, float, and so forth.
   d. Detailed procedures for preoperational checks such as overspeed, trip and throttle valve operation, and

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valve lifting sequence including setting and adjustment.

e. Coupling installation procedures.

6.2.6 RECOMMENDED SPARES

The vendor shall submit a supplementary proposal for spare parts other than those included in the vendor's original quotation. This supplementary proposal shall include recommended spare parts, cross-sectional or assembly-type drawings, part numbers, materials, prices, and delivery. Part numbers shall identify each part for interchangeability purposes. Standard purchased items shall be identified by the original manufacturer's numbers. This supplementary proposal shall be forwarded to the purchaser promptly after receipt of approved drawings and in time to permit ordering and delivery of parts prior to field start-up. The transmittal letter shall be identified by data in 6.2.1.2.

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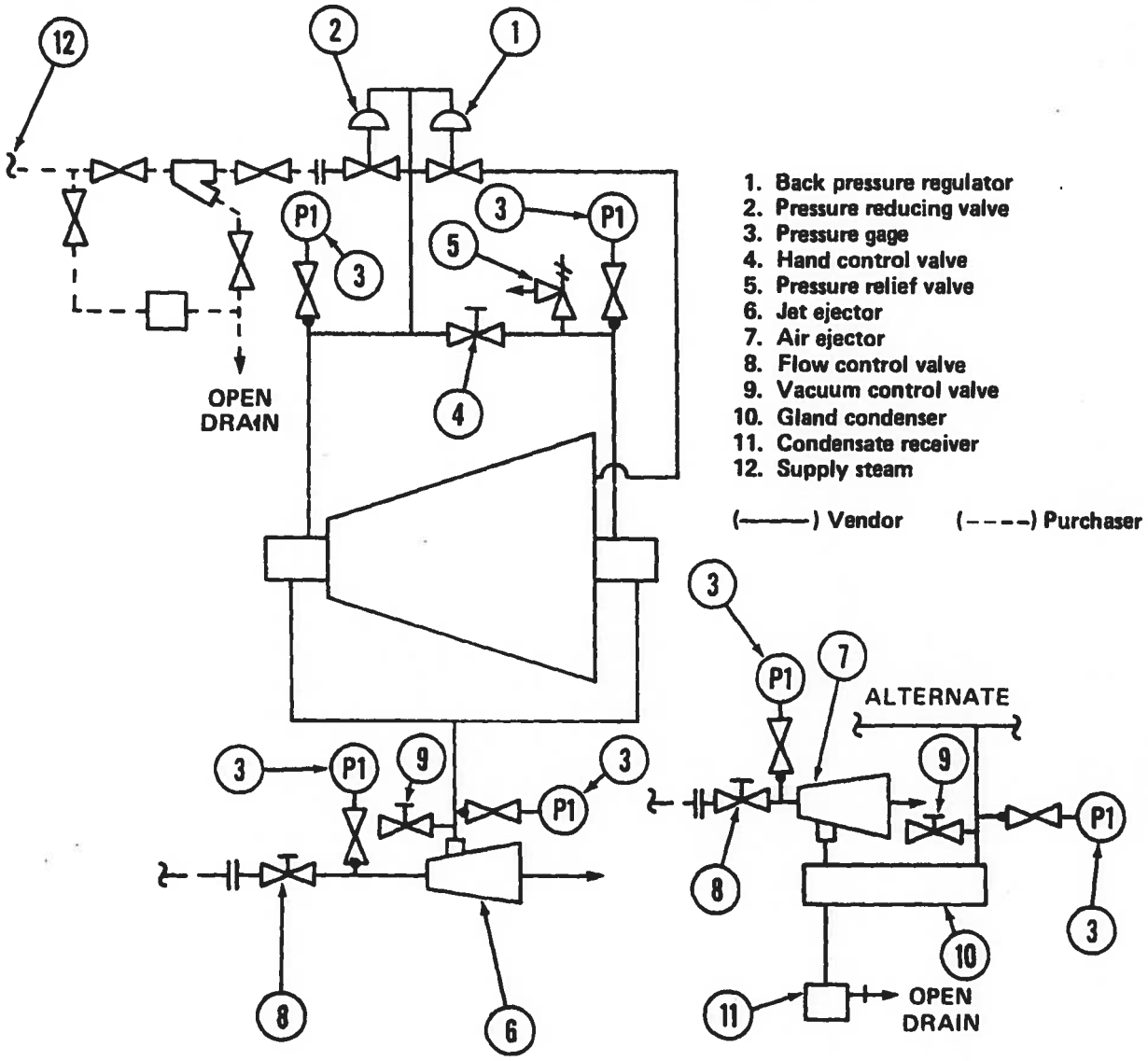
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APPENDIX A

GLAND SEALING AND LEAK-OFF EQUIPMENT



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APPENDIX B

RATED LOAD CURVES FOR STANDARD 6 x 6 SHOE THRUST BEARINGS

Approximately 50 percent of ultimate load

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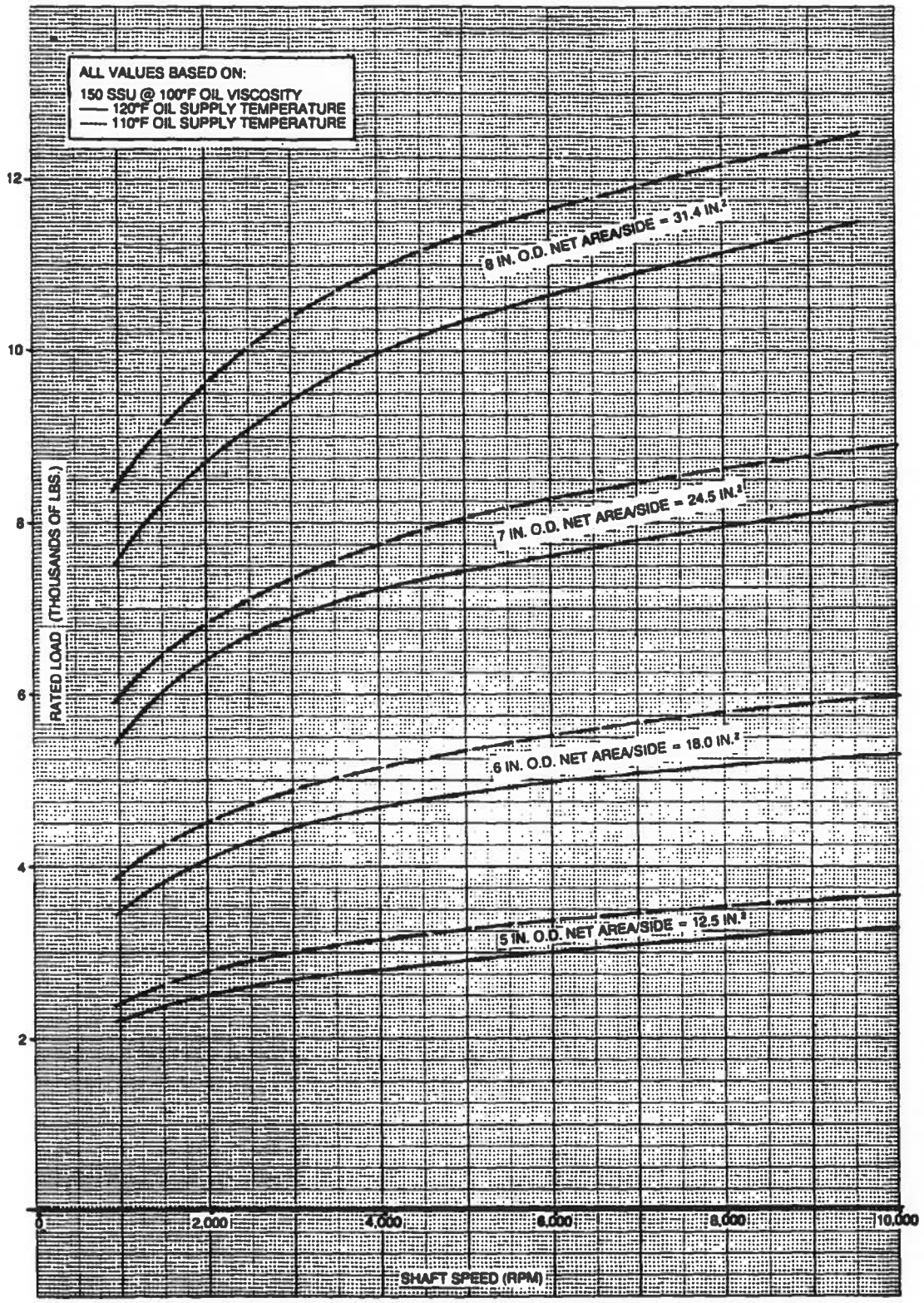


Figure B-1 — Rated load versus shaft speed for bearings of 5-, 6-, 7-, and 8-inch OD based on oil supply temperatures of 110F and 120F and an oil viscosity of 150 SSU at 100F

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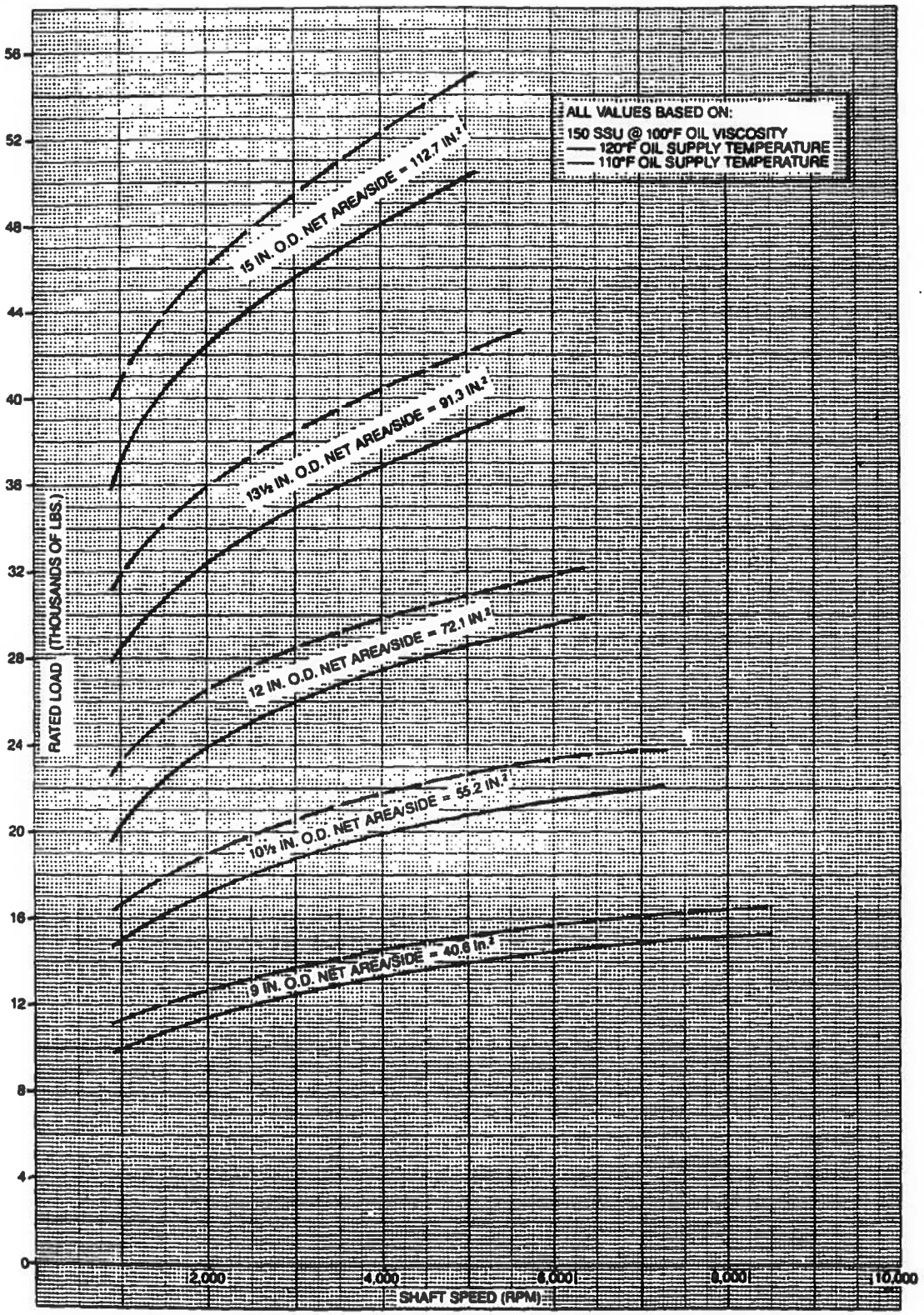


Figure B-2 — Rated load versus shaft speed for bearings of 9-. 10¼-. 12. 13¼-, and 15-inch OD based on oil supply temperatures of 110F and 120F and an oil viscosity of 150 SSU at 100F

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APPENDIX C

TYPICAL DATA SHEETS

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CONTRACT NO. _____
ITEM NO. _____
REV _____ DATE _____
BY _____ REVIEWED _____
SHEET _____ OF _____
P.O. NO _____

DATA SHEET

SPECIAL-PURPOSE STEAM TURBINE

APPLICABLE TO: ○ PROPOSAL ○ PURCHASE ○ AS BUILT
FOR _____ UNIT _____
SITE _____ MODEL NO. _____
SERVICE _____ TYPE _____
MANUFACTURER _____ NO. REQUIRED _____ SERIAL NO. _____

NOTE: ○ Indicates information to be Completed by PURCHASER ☐ by MANUFACTURER

OPERATING CONDITIONS

INDICATE GUARANTEE POINT BY *

	POWER, Hp	SPEED, RPM	STEAM RATE LB/Hp-HR
☐ Rated			
☐ Normal			
☐			

☐ Steam Rates Based on Output Shaft of: ☐ Turb. ☐ Gear
Exhaust Enthalpy @ Rated Point _____ BTU/LB

SPEEDS:
Criticals: 1st _____ RPM 2nd _____ RPM Trip _____ RPM
Max. Continuous _____ RPM

STEAM CONDITIONS:

○ INLET STEAM:
Rated (Normal) _____ PSIG _____ °FTT
Maximum _____ PSIG _____ °FTT
Minimum _____ PSIG _____ °FTT

○ EXHAUST STEAM:
Rated (Normal) _____ PSIG/In. HgA _____ °FTT
Maximum _____ PSIG/In. HgA _____ °FTT
Minimum _____ PSIG/In. HgA _____ °FTT

○ EXTRACTION STEAM: ___Controlled ___Uncontrolled
Rated (Normal) _____ PSIG _____ °FTT
Maximum _____ PSIG _____ °FTT
Minimum _____ PSIG _____ °FTT

Flow (lb/hr) _____ Normal _____ Max. _____ Min

○ ADMISSION STEAM: ___Controlled ___ Uncontrolled
Rated (Normal) _____ PSIG _____ °FTT
Maximum _____ PSIG _____ °FTT
Minimum _____ PSIG _____ °FTT

Flow (lb/hr) _____ Normal _____ Max. _____ Min.

☐ Max. Throttle Flow _____ lb/hr
☐ Max. Flow to Condenser _____ lb/hr @ _____ In. HgA
☐ Max. Exhaust Temp. In Operation _____ °F
☐ Max. BHP Required @ Min./ Inlet Max. Exh. _____ Bhp

DUTY:
○ Continuous ○ Intermittent ○ Auto Start

LOCATION:
○ Indoor ○ Heated ○ Under Roof
○ Outdoor ○ Unheated ○ Partial Sides
○ Grade ○ Mezzanine
○ Winterization Req'd. ○ Tropicalization Req'd.

SITE DATA:
○ Elevation _____ Ft. ○ Bar. _____ PSIA/In.HgA
○ Temperature: _____ °F Summer _____ °F Winter
○ Relative Humidity _____ % Design Wet Bulb _____ °F

UNUSUAL CONDITIONS:
○ Dust ○ Fumes _____
○ Other _____

CONSTRUCTION FEATURES

TURBINE TYPE:
○ Horizontal ○ Vertical ☐ Single-Valve ☐ Multi-Valve
☐ No. Stages: _____ Impulse _____ Reaction
☐ Overspeed Trip Device ___ Mech. _____ Elect. _____ Hyd.
○ Casing Split: _____ Horizontal _____ Vertical
☐ Rotor: _____ Solid _____ Built-Up _____ Combination
☐ Rotation (Facing Gov. End.): _____ CW _____ CCW
☐ Exhaust Flow _____ Single _____ Double

GOVERNOR TYPE:
○ Electronic ○ Hydraulic ○ Oil Relay ○ Direct Acting
○ NEMA Class _____
○ Governor MFR _____ ☐ Model _____
☐ Synch. Motor _____ HP ☐ Oil Cooler ☐ Oil Heater
☐ Power Cylinder ○ Governor Purge Required
○ Stn. Stl. Pins & Bushings in Gov-Linkage
○ Isochronous Control ○ Speed Droop Control
☐ Speed Changer _____ Manual _____ Pneumatic _____ Electronic
☐ Speed Range _____ RPM @ _____ PSIG/mA to
_____ RPM @ _____ PSIG/mA

VALVES:

	Cam Lift	Bar Lift	Grid Valve
☐ No. Auto Gov. Valves _____	___	___	___
☐ No. Auto Extr. Valves _____	___	___	___
☐ No. Auto Adm. Valves _____	___	___	___

○ Separate Trip & Throttle Valve: ___ Hyd. ___ Mech.
___ Remote Trip ___ Manual Actuation
___ Spring Support Req'd ___ Manual Excerciser
○ Extraction Non-Return Valve: ___ Hyd ___ Mech
___ Remote Trip ___ Manual Actuation
○ Admission Stop Valve: ___ Hyd ___ Mech.
___ Remote Trip ___ Manual Actuation
○ Auto Valves arranged to close on Trip _____ Yes _____ No

BLADES (BUCKETS):
☐ Max. Tip Speed _____ FPS
☐ Final Stage Blade Length _____ in. Max Length _____ in.
☐ Nozzle Ring: _____ % Admission ___ Welded ___ Removable
☐ Blade Root: ___ Dovetail ___ "T" ___ Fir Tree
☐ Shrouds: ___ Welded ___ Riveted ___ Wire Damp.

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CONTRACT NO. _____
ITEM NO. _____
REV. _____ DATE _____
BY _____ REVIEWED _____
SHEET _____ OF _____

DATA SHEET
SPECIAL-PURPOSE STEAM TURBINE

CONSTRUCTION FEATURES

BEARING HOUSING CONSTRUCTION:
- ☐ Type (Separate, Integral) _____ Split _____
- ☐ Material _____

RADIAL BEARINGS:
- ☐ Type _____ Brg. Span (In.) _____
- ☐ Area (In^2) _____ Loading (PSI): Act. _____ Allow. _____

THRUST BEARING:
- ☐ Location _____ Type _____
- ☐ MFR _____ Area ($In.^2$) _____
- ☐ Loading PSI Act. _____ Allow. _____

PACKING TYPE:
- ☐ Interstage Gland Seals _____
- ☐ End-Gland Seals _____ No. Per Box _____
- ☐ ΔP Per Seal _____ PSI ☐ Material _____
- ○ Insulation _____ ○ Jacket Material _____
- ☐ Total Expected Leakage _____ Lb/Hr _____
- ☐ Labyrinth Type _____ Stationary _____ Rotating

GLAND SEALING SYSTEM ○ See Separate Data Sheet Attached
- ○ Gland Condenser w/Mounting Feet Req'd.
- ○ Steam Ejector Req'd. _____ PSIG _____ °FTT
- ○ Vacuum Device Req'd. ○ Compound Press. Gauge Req'd.
- ○ Auto. Pressure Control Valve (Motive Steam) (Sealing Steam)
- ○ Gld. Cond. Drain: _____ Loop Seal _____ Vertical Leg _____ Other _____
- ☐ Minimum Cond. Drain Height _____ In.
- ○ Relief Valve on Sealing Steam Header

CAPABILITIES:
- ☐ Max. Steam Thru Inlet Valve _____ lb/hr*
- ☐ Max. Steam Thru Exhaust Valve _____ lb/hr*
- ☐ Steam Inlet Parts:
  - Max. Allow. Press. _____ PSIG Temp. _____ °F
- ☐ Exhaust End Casing:
  - Max. Allow Pressure _____ PSIG/In. HgA
  - Max. Allow. Temperature _____ °F
  - Max. Expected Temp. During No-Load Run-In _____ °F
- ☐ Sentinel Warning Valve Set @ _____ PSIG
- ☐ Hydrostatic Test Pressure:
  - Steam Inlet Parts _____ PSIG
  - Exhaust End _____ PSIG/In. HgA
- ☐ Future Potential Maximum Horsepower _____ BHP*

*With Minor Modifications.

TURBINE MATERIALS:
- ☐ High Press. Casing _____
- ☐ Exhaust Casing _____ ☐ Mid Casing _____
- ☐ Nozzles _____ ☐ Blades _____
- ☐ Wheel _____ ☐ Shaft _____
- ☐ Steam Chest _____
- ☐ Diaphrams _____
- ☐ Nozzle Rings _____
- ☐ Shrouds _____
- ☐ Gov Valve Trim _____
- ☐ Labyrinth Seals _____
- ☐ Shaft Materials under Seals _____
- ☐ _____ Applied By _____ Spraying _____ Plating _____ Sleeves

MAIN CONNECTIONS:

	SIZE	ANSI RATING	FACING	POSITION
☐ Inlet				
☐ Exhaust				
☐ Extraction				
☐ Admission				

ALLOWABLE PIPING FORCES AND MOMENTS:

	INLET FORCE LB	INLET MOM'T FT.-LB	EXHAUST FORCE LB	EXHAUST MOM'T FT LB	EXTR./ADM. FORCE LB	EXTR./ADM. MOM'T FT.-LB
☐ Parallel to Shaft						
☐ Vertical						
☐ Horiz. 90°						

OTHER CONNECTIONS:

SERVICE	NO.	SIZE	ANSI RATING	FACING
☐ Lube Oil Inlet				
☐ Lube Oil Outlet				
☐ Gland Condenser Conn				
☐ Staging Drains				
☐ Steam Ring Drain				
☐ T&T Valve H.P. Stem Leakoff				
☐ T&T Valve L.P. Stem Leakoff				
☐ T&T Valve Above Seat Drain				
☐ T&T Valve Below Seat Drain				
☐ Cooling Water Conn.				
☐ Purge for Brg Housing				
☐ Exhaust Case				

BASEPLATES & SOLEPLATES:
- ○ Baseplate By _____
- ○ Decked with Non-Skid Deck Plate ○ Open Construction
- ○ Drip Rim _____ With Open Drain _____
- ○ Under Turbine Only _____ Other _____
- ○ Soleplate By _____
- ○ Horiz. Adjusting Screws for Equipment _____
- ○ Furnish SS Shims
- ○ Foundation Bolts Furn. By: _____ Vendor _____ Purchaser
- ☐ Leveling (Chock) Blocks Req'd.

REFERENCE SPECIFICATIONS:

API 612 SPECIAL PURPOSE STEAM TURBINE

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ITEM NO. _____
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SHEET _____ OF _____

DATA SHEET
SPECIAL-PURPOSE STEAM TURBINE

UTILITIES

COOLING WATER:

	NORM	MAX	DESIGN
○ Supply Pressure, PSIG			
○ Supply Temp., °F			
○ Pressure Drop PSI			

○ Max. Temperature Rise Allowed ____ °F
☐ Quantity Required
____GPM For _____
____GPM For _____

AUXILIARY STEAM SUPPLY:

○ Supply Pressure, PSIG
Normal ____ Max ____ Min ____
○ Temperature ____ °F Nor ____ Max ____ Min ____

☐ Quantity Required
____Lbs/Hr For _____
____Lbs/Hr For _____

INSTRUMENT AIR SUPPLY:

○ Pressure PSIG _____ Max. _____ Min

AUXILIARY MOTORS:

○ _____ Volts _____ Phase _____ Hertz

INSTRUMENTATION

Gauge Readout In ☐ English ☐ SI ☐ Dual ☐ Other _____

NOTE: ☐ SUPPLIED BY VENDOR ○ SUPPLIED BY PURCHASER Δ LOCATED ON A MACHINE MOUNTED INSTRUMENT BOARD

PRESSURE GAGE REQUIREMENTS:

FUNCTION	LOCALLY MOUNTED	LOCAL PANEL	FUNCTION	LOCALLY MOUNTED	LOCAL PANEL
Lube Oil Pump Discharge _____	☐○Δ	☐○	1st Stage Steam _____	☐○Δ	☐○
Lube Oil Filter ΔP _____	☐○Δ	☐○	Steam Chest _____	☐○Δ	☐○
Lube Oil Supply _____	☐○Δ	☐○	Exhaust Steam _____	☐○Δ	☐○
Gov. Control Oil _____	☐○Δ	☐○	Extraction Steam _____	☐○Δ	☐○
Gov Control Oil ΔP _____	☐○Δ	☐○	Steam Ejector _____	☐○Δ	☐○
Coupling Oil ΔP _____	☐○Δ	☐○	Steam Seal _____	☐○Δ	☐○
Main Steam Inlet _____	☐○Δ	☐○	_____	☐○Δ	☐○

TEMPERATURE GAGE REQUIREMENTS:

FUNCTION	LOCALLY MOUNTED	LOCAL PANEL	FUNCTION	LOCALLY MOUNTED	LOCAL PANEL
Lube Oil Discharge From Each	☐○Δ	☐○	Cooler Oil Inlet & Outlet _____	☐○Δ	☐○
Turbine Journal Bearing	☐○Δ	☐○	Steam Inlet _____	☐○Δ	☐○
Turbine Thrust Bearing _____	☐○Δ	☐○	Steam Exhaust _____	☐○Δ	☐○
Gear Journal Bearing _____	☐○Δ	☐○	_____	☐○Δ	☐○

ALARM & SHUTDOWN FUNCTIONS:

FUNCTION	PRE ALARM	TRIP	FUNCTION	PRE ALARM	TRIP
☐○ Low Lube Oil Pressure Ea. Level			☐○ Turbine Vibration		
☐○ Hi Oil Filter ΔP			☐○ Turbine Axial Position		
☐○ Aux. Lube Oil Pump Start			☐○ Trip & Throttle Valve Position		
☐○ Hi Oil Cooler Outlet Temp.			☐○ Hi Turb. Steam Seal Leakage		
☐○ Low Control Oil Pressure			☐○ Hi Turb. Exh. Pressure		
☐○			☐○ Hi Turb. Ext. Pressure		
☐○			☐○ Turb. Overspeed Trip Operation		

MISCELLANEOUS INSTRUMENTATION:

☐○ Turbine Speed Pick-Up Devices ☐ Electronic ☐ Other _____
☐○ Turbine Speed Indicators
Turbine Speed Indicators Located On. ☐ Local Panel ☐ Main Board Type ☐ Digital ☐ Dial Gauge
☐○ Remote Hand Speed Changer – Mounted on Local Panel
☐○ Alarm Horn & Acknowledgement Switch on Panel (Local) (Main)
☐○ _____
☐○ _____

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SHEET _____ OF _____

DATA SHEET
SPECIAL-PURPOSE STEAM TURBINE

COUPLING:
- ○ Mount ½ Coupling ○ Type _____
- ☐ MFR _____
- ☐ Spacer Req'd _____
- ○ Coupling Furnished by _____
- ☐ Keyed (1) or (2); or Hydr. Fit _____
- ☐ Cplg. Coeff. Frict. _____ Cplg. Gear Pitch Dia. _____ in.

Turbine Shaft: ___Taper _____Cylindrical
Generator Shaft: ___Taper _____Cylindrical
Coupling Guard: ___Mfr. Std. _____Other
- ☐ Lubrication Req'd. _____Yes _____No

COMPOSITE TORSIONAL ANALYSIS:
- ○ Required ○ By _____

LATERAL CRITICAL SPEED ANALYSIS:
- ○ Required ○ By _____

SHOP INSPECTION AND TESTS:

	REQ'D	WITNESS
○ Shop Inspection	○	○
○ Hydrostatic (Piping, Casing, Condenser)	○	○
○ Blade Shaker (Static) — Rows	○	○
○ Rotor Balance: ___Vacuum Pit ___Std.	○	○
○ Rotor Thermal Stability	○	○
○ Mechanical Run: ___w/Job Cplg. ½	○	○
○ Run Spare Rotor ___w/Job Cplg. ½	○	○
○ Performance Test: _Part Load _Full Load	○	○
○ Functional Test: ___T & T Valve	○	○
___Turning-Gear	○	○
___Gland Condenser	○	○
___Sealing System	○	○
___ Lube Oil System	○	○
○ Governor Stability	○	○
○ Turb. Run at Overspeed and Trip	○	○
○ Use Shop Lube System	○	○
○ Use Job Lube System	○	○
○ Use Job Vib & Axial Disp. Probes, Oscillator-Demodulators & Monitor	○	○
○ Use Shop Vibration Probes etc.	○	○
○ X–Y–Y' Plot	○	○
○ Check Brg. & Seals after Test	○	○
○ Noise Level Test	○	○
○ Preparation for Shipment	○	○
○ FM Tape Recording (Vibration)	○	○
○ Cleanliness Inspection (Para. 4.2.6)	○	○
○ Hardness Tests (Para. 4.2.7.)	○	○

OIL REQUIREMENTS:

	CONTROL OIL	LUBE OIL
☐ Max. Quantity, GPM		
☐ Pressure, PSIG		
☐ Temperature, °F		
☐ Total Heat Reject BTU/Hr		

- ○ Oil Req'd for ____T&T Valve _____Hyd Non-Return Valve
- ☐ Oil Viscosity _____ SSU @ 100 °F

VIBRATION DETECTORS:
- ○ Type _____ ☐ Model _____
- ○ Mfr. _____
- ○ No. at Each Shaft Bearing _____ Total No. _____
- ○ Oscillator-Demodulators Supplied by _____
  - ○ Mfr. _____ ☐ Model _____
- ○ Monitor Supplied by _____ No. Channels _____
  - ○ Location _____ Enclosure _____
  - ○ Mfr. _____ ☐ Model _____
  - ☐ Scale Range _____ ○ Alarm: ☐ Set @ _____ Mils
  - ○ Shutdown: ☐ Set @ ___ Mils ○ Time Delay _____ Sec.

AXIAL MOVEMENT DETECTORS:
- ○ Type _____ ☐ Model _____
- ○ Mfr. _____ No. Required _____
- ○ Oscillator-Demodulators Supplied by _____
  - ○ Mfr. _____ ☐ Model _____
- ○ Monitor Supplied by _____ No. Channels _____
  - ○ Location _____ Enclosure _____
  - ○ Mfr. _____ ☐ Model _____
  - ☐ Scale Range _____ ○ Alarm: ☐ Set @ _____ Mils
  - ○ Shutdown: ☐ Set @ _____ Mils ○ Time Delay _____ Sec.

KEYPHAZOR SENSOR:
- ○ Type _____ ☐ Model _____
- ○ Mfr. _____ ○ No. Req'd _____
- ○ Location _____
- ○ Oscillator-Demodulators Supplied by _____
  - ○ Mfr _____ ☐ Model _____
- ○ Monitor Supplied by _____ No. Channels _____
  - ○ Location _____ Enclosure _____
  - ○ Mfr _____ ☐ Model _____

BEARING TEMPERATURE DEVICES:
- ○ Thermistors:
  - ○ Type: __Pos. Temp. Coeff. __ Neg. Temp. Coeff
  - ○ Temp. Switch & Indicator By: __ Purch. __ Mfr
- ○ Thermocouples:
  - ○ Selector Switch & Ind. By: __ Purch. __ Mfr.
- ○ Resistance Temp. Detectors:
  - ○ Resistance Mat'l _____ ☐ _____ OHMS
  - ○ Selector Switch & Ind. By: ___ Purch. ___ Mfr.
- ○ LOCATION – JOURNAL BRG:

  No.: _Ea. Pad _Every Other Pad _Per Brg.
  _ Other _____
- ○ LOCATION – THRUST BRG:

  No. (Act.): _Ea. Pad _ Every Other Pad _ Per Brg.
  _ Other _____
  No. (Inact.): _Ea. Pad _Every Other Pad _ Per Brg.
  _Other _____
- ☐ Max. Bearing Temps.:

  _____ °F for Alarm _____ °F for Shutdown

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ITEM NO. _____
REV. _____ DATE _____
BY _____ REVIEWED _____
SHEET _____ OF _____

DATA SHEET
SPECIAL PURPOSE STEAM TURBINE

INSTRUMENTATION

EXHAUST RELIEF VALVE REQUIREMENTS:
- ○ Max Set Pressure _____ PSIG Exhaust.
- ○ Steam Flow _____ Lb/Hr
- ○ Supplied By _____ Purchaser _____ Vendor

EXTRACTION RELIEF VALVE REQUIREMENTS:
- ○ Max Set Pressure _____ PSIG Extraction
- ○ Steam Flow _____ Lb/Hr
- ○ Supplied By _____ Purchaser _____ Vendor

AREA CLASSIFICATION
Class _____ Group _____ Division _____

MOTOR CONTROL & INSTRUMENT VOLTAGE
_____ Volts _____ Phase _____ Cycles

ALARM & SHUTDOWN VOLTAGE.
_____ Volts _____ Phase _____ Cycles or _____ DC
Solenoid Voltage _____

LOCAL CONTROL PANEL:
Furnished By; ☐ Vendor ☐ Purchaser ☐ Others _____
☐ Free Standing ☐ Weatherproof ☐ Totally Enclosed ☐ Extra Cutouts
☐ Vibration Isolators ☐ Strip Heaters ☐ Purge Connections ☐ With Doors
☐ Annunciator: Furnished By: ☐ Vendor ☐ Purchaser ☐ Others
Annunciator located on ☐ Local Panel ☐ Main Control Board
☐ Customer Connections Brought out to Terminal Boxes By Vendor
Remarks: _____

INSTRUMENT SUPPLIERS:

Pressure Gages:	MFR. _____	Size & Type _____
Temperature Gages:	MFR. _____	Size & Type _____
Level Gages:	MFR. _____	Size & Type _____
Diff. Pressure Gages:	MFR. _____	Size & Type _____
Pressure Switches:	MFR. _____	Size & Type _____
Diff. Pressure Switches:	MFR. _____	Size & Type _____
Temperature Switches:	MFR. _____	Size & Type _____
Level Switches:	MFR. _____	Size & Type _____
Control Valves:	MFR. _____	Size & Type _____
Pressure Relief Valves:	MFR. _____	Size & Type _____
Thermal Relief Valves:	MFR. _____	Size & Type _____
Sight Flow Indicators:	MFR. _____	Size & Type _____
Pneu. Pressure Transmitters:	MFR. _____	Size & Type _____
Vibration Equipment:	MFR. _____	Size & Type _____
Tachometer:	MFR. _____	Range & Type _____
Solenoid Valves:	MFR. _____	Size & Type _____
Annunciator:	MFR. _____	Model & No. Points _____
Thermocouples:	MFR. _____	Size & Type _____
Resistance Temp. Detectors:	MFR. _____	Size & Type _____
Thermowells:	MFR. _____	Size & Type _____
	MFR. _____	Size & Type _____
	MFR. _____	Size & Type _____
	MFR. _____	Size & Type _____

SWITCHES:
Enclosures ○ Explosion Proof ○ Weather Proof ○ Other _____
Alarm Contacts Shall: ☐ Open ☐ Close to Sound Alarm and be Normally ☐ Energized ☐ De-Energized
Shutdown Contacts Shall: ☐ Open ☐ Close to Trip and be Normally ☐ Energized ☐ De-Energized
Note: Normal Condition is when Turbine is in Operation

MISCELLANEOUS:
Pre-Alarm and Shutdown Switches shall be Separate.
Purchasers Electrical and Instrument Connections within the confines of the Baseplate and Console shall be:
☐ Brought out to Terminal Boxes. ☐ Made Directly by the Purchaser.
Comments Regarding Instrumentation: _____

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DATA SHEET
SPECIAL-PURPOSE STEAM TURBINE

GEAR ○ Separate Data Sheet Attached
○ Special Purpose Gear Req'd.
○ Gear Furnished By _____

OIL SYSTEM
Furnished By ○ Turbine Mfr. ☐ Others
○ Separate for Turbine Only
○ Common with Driven Equipment
Turbine Mfr. to Supply
○ Stainless Steel Oil Supply Header Piping
○ Oil Drain Header Piping
○ Stainless Steel ○ Carbon Steel
_____
_____

EMERGENCY TURNING-GEAR
○ Turning Gear Req'd. ○ Quick Start Required
☐ Mfr. _____ ☐ Model _____
☐ Ratio _____
☐ Motor: _____ Electric _____ HP. _____ Volts ○ AC ○ DC
_____ Pneumatic _____ PSIG _____ °F _____ SCFM
_____ Auto Engage _____ Manual Engage

PAINTING
○ Manufacturer's Standard
○ Other _____

SHIPMENT
○ Domestic ○ Export
○ Export Boxing Required ○ Outdoor Storage over 6 months
○ Water Proof Boxing Required
○ Spare Rotor Assembly Packaged for
○ Horizontal Storage ○ Vertical Storage

WEIGHTS:
☐ Turbine _____ lb.
☐ Rotor _____ lb.
☐ Turb. Upper ½ Casing _____ lb.
☐ Maximum for Maintenance (Identify) _____
_____ lb.
☐ Total Shipping Weight _____ lb.

SPACE REQUIREMENTS:
☐ Complete Unit: L _____ in. W _____ in. H _____ in.
☐ Control Panel: L _____ in. W _____ in. H _____ in.
_____

MISCELLANEOUS:
○ Provisions for Field Balancing
○ Vendor's Review and Comment on Purchaser's Piping and Foundation Drawings are Required
○ Shaft Grounding Devices
○ "Y" Type Strainer
○ Water Washing Connections
○ Optical Alignment Plate
○ Insulation (Lagging) Req'd
○ Jacket Req'd
○ Axial Alignment Key
○ Blade Diagrams: ___ Campbell ___ Goodman
___ Soderberg ___ Other _____
○ Metric/English
_____ Drawings _____ External Flanges
_____ Internal Bolting and Threads
_____

SKETCH:

REMARKS: _____

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DATA SHEET

TURBINE GLAND CONDENSER

ITEM NO. _____ DUTY _____ BTU/HR
SUPPLIER _____ NO. UNITS REQUIRED _____
MODEL NO. _____ TYPE _____

NOTE: ○ INDICATES INFORMATION TO BE COMPLETED BY PURCHASER ☐ BY MANUFACTURER

OPERATING CONDITIONS

	SHELL SIDE	TUBE SIDE
○ Fluid		
☐ Total Flow LBS/HR		
☐ Specific Gravity	@ °F	@ °F
☐ Thermal Cond. BTU/HR. x SQ. FT. x °F	@ °F	@ °F
☐ Specific Heat – BTU/LB. x °F	@ °F	@ °F
☐ Viscosity – SSU	@ °F	@ °F
☐ Operating Temperatures, °F	IN OUT	IN OUT
Inlet Pressure, PSIG		
Inlet Velocity, FT/SEC		
○ Pressure Drop, PSI	ALLOW. CALC.	ALLOW. CALC.
☐ Design Temperature, °F		
Pressure, PSIG	MIN. TEST	MIN. TEST
Foul Resistance, SQ. FT. x HR. x °F/BTU		
○ Min Corrosion Allowance, in.		
☐ Number of Passes Per Shell		

CONSTRUCTION DETAILS

☐ Total Area (1), Sq. Ft. _____
☐ LMTD _____
☐ Corrected MTD _____
☐ Transfer Rate, Clean _____
☐ Transfer Rate, Service _____
☐ Cross Baffles, Type _____
○ Code Requirements (2) ___ ASME; TEMA _____

☐ Shell, No. x ID. _____ x _____ IN
☐ Tubes, No. Per Shell _____
☐ O.D. x Length _____ IN. x _____ IN.
☐ Gauge, BWG _____ Avg., Min. Wall
☐ Tube Pitch _____ IN. △ □ ◇
☐ Removable Tube Bundle ☐ YES ☐ NO
○ Code Stamp ☐ YES ☐ NO

☐ Weights Es. Bundle _____ Lbs.; Bundle & Shell _____ Lbs.; Full of Water _____ lbs.

☐ NOZZLE SIZES

	SHELL SIDE			TUBE SIDE		
	NO.	SIZE	RATING & FACING	NO.	SIZE	RATING & FACING
INLET						
OUTLET						
DRAIN						
VENT						

○ MATERIALS

Tubes _____ Baffles _____
Tube Sheets _____ Channel _____
Shell _____ Channel Flanges _____
Shell Flanges _____ Channel Noz. Flanges _____

REMARKS: _____

(1) OUTSIDE TUBE AREA EXCLUDING AREA IN TUBE SHEETS.
(2) UNITS EXEMPT FROM CODE STAMP SHALL HAVE LONGETUDINAL WELD SEAMS SPOT EXAMINED PER PARA UW-52 OF ASME CODI

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APPENDIX D

SPECIAL-PURPOSE STEAM TURBINE VENDOR DRAWING AND DATA REQUIREMENTS

This appendix consists of a sample distribution record (schedule), followed by a description of the items that are presented numerically in the schedule.

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Job No. _____ Item No. _____
Purchase Order No. _____
Requisition No. _____
Inquiry No. _____
Location _____ Unit _____ Date _____
Service _____ By _____

Proposal — Bidder shall furnish _____ copies of drawings and data indicated

Review — Vendor shall furnish _____ copies and _____ transparencies of drawings and data indicated. Vendor shall furnish _____ operating and maintenance manuals.

Final

DISTRIBUTION RECORD

			DESCRIPTION	Final — Received from vendor	Final — Due from vendor *	Review — Returned to vendor	Review — Received from vendor	Review — Due from vendor *		
			1. CERTIFIED DIMENSIONAL OUTLINE DRAWING AND LIST OF CONNECTIONS							
			2. CROSS-SECTIONAL DRAWING AND BILL OF MATERIALS							
			3. ROTOR ASSEMBLY DRAWING AND BILL OF MATERIALS							
			4. THRUST-BEARING ASSEMBLY DRAWING AND BILL OF MATERIALS							
			5. JOURNAL-BEARING ASSEMBLY DRAWING AND BILL OF MATERIALS							
			6. PACKING AND LABYRINTH DRAWINGS AND BILL OF MATERIALS							
			7. COUPLING ASSEMBLY DRAWING AND BILL OF MATERIALS							
			8. GLAND SEALING AND LEAKOFF SCHEMATIC AND BILL OF MATERIALS							
			9. GLAND SEALING AND LEAKOFF ARRANGEMENT DRAWING AND LIST OF CONNECTIONS							
			10. GLAND SEALING AND LEAKOFF COMPONENT DRAWINGS AND DATA							
			11. LUBE-OIL SCHEMATIC AND BILL OF MATERIALS							
			12. LUBE-OIL ARRANGEMENT DRAWING AND LIST OF CONNECTIONS							
			13. LUBE-OIL COMPONENT DRAWINGS AND DATA							
			14. ELECTRICAL AND INSTRUMENTATION SCHEMATICS AND BILL OF MATERIALS							
			15. ELECTRICAL AND INSTRUMENTATION ARRANGEMENT DRAWING AND LIST OF CONNECTIONS							
			16. CONTROL AND TRIP SYSTEM							
			17. GOVERNOR DETAILS							
			18. STEAM FLOW VERSUS HORSEPOWER							
			19. STEAM FLOW VERSUS FIRST-STAGE PRESSURE							
			20. STEAM FLOW VERSUS SPEED AND EFFICIENCY							
			21. STEAM FLOW VERSUS THRUST-BEARING LOAD							
			22. EXTRACTION PERFORMANCE CURVES							
			23. STEAM CORRECTION CHARTS							
			24. VIBRATION ANALYSIS DATA							
			25. LATERAL CRITICAL ANALYSIS							
			26. ALLOWABLE FLANGE LOADING							
			27. ALIGNMENT DIAGRAM							
			28. WELD PROCEDURES							
			29. HYDROSTATIC TEST LOGS							
			30. MECHANICAL RUN TEST LOGS							
			31. ROTOR BALANCE LOGS							
			32. ROTOR MECHANICAL AND ELECTRICAL RUNOUT							
			33. "AS-BUILT" DATA SHEETS							
			34. "AS-BUILT" DIMENSIONS							
			35. OPERATING AND MAINTENANCE MANUALS							
			36. SPARE PARTS RECOMMENDATION AND PRICE LIST							

* Bidder shall complete these two columns to reflect his actual distribution schedule and include this form with his proposal.

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NOTES:

1. Vendor shall proceed with manufacture upon receipt of order and without awaiting purchaser's approval of drawings where necessary to meet scheduled shipping date.
2. Send all drawings and data to _____
   _____
3. All drawings and data must show project, appropriation, purchase order, and item numbers in addition to plant location and unit. One set of drawings/instructions necessary for field installation must be forwarded with shipment in addition to copies specified above.
4. See Description below.
5. All of the requested information indicated above shall be received before final payment will be made.

Nomenclature:

wpis — weeks prior to shipment
wafo — weeks after firm order
waad — weeks after receipt of approved drawings

Vendor _____

Date _____ Vendor Reference _____

Signature _____
(Signature acknowledges receipt of all instructions)

DESCRIPTION

1. Certified dimensional outline drawing including—
   a. Size, rating, and location of all customer connections.
   b. Approximate overall and handling weights.
   c. Overall dimensions.
   d. Shaft centerline height.
   e. Dimensions of baseplates (if furnished) complete with diameter, number, and locations of bolt holes and thickness of metal through which bolts must pass.
2. Cross-sectional drawings and bill of materials including—
   a. Journal-bearing clearances and tolerance.
   b. Rotor float (axial).
   c. Seal clearances (shaft end and internal labyrinth) and tolerance.
   d. Axial position of wheel(s) relative to inlet nozzle or diaphragms, and tolerance allowed.
   e. OD of all wheel(s) @ blade tip.
3. Rotor assembly drawing including—
   a. Axial position from active thrust collar face to—
      1. Each wheel (inlet side)
      2. Each radial probe
      3. Each journal bearing centerline.
      4. Phase angle notch.

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b. Thrust collar assembly details including—
   1. Collar-shaft fit with tolerance
   2. Concentricity (or axial runout) tolerance
   3. Required torque for locknut
   4. Surface finish requirements for collar faces
   5. Preheat method and temperature requirements for "shrunk-on" collar installation.
c. Dimensioned shaft end(s) for coupling mounting(s).
d. Bill of materials.

4. Thrust-bearing assembly drawing and bill of materials.

5. Journal-bearing assembly drawing and bill of materials.

6. Packing or labyrinth drawings and bill of materials.

7. Coupling assembly drawing and bill of materials.

8. Gland-sealing and leakoff schematic including—
   a. Steady-state and transient steam and air flows and pressures.
   b. Relief and control valve settings.
   c. Utility requirements including electrical, water, steam, and air.
   d. Pipe and valve sizes.
   e. Instrumentation, safety devices, and control schemes.
   f. Bill of materials.

9. Gland-sealing and leakoff arrangement drawing including size, rating, and location of all customer connections.

10. Gland-sealing and leakoff component outline and sectional drawings and data including—
    a. Gland condenser fabrication drawing and bill of materials.
    b. Completed data form for condenser.
    c. Air or water ejector drawing and performance curves.
    d. Control valves, relief valves, and instrumentation.
    e. Vacuum pump schematic, performance curves, cross section, outline drawing, and utility requirements (if pump is furnished).

11. Lube oil schematic including—
    a. Steady-state and transient oil flows and pressures at each use point.
    b. Control, alarm, and trip settings (pressure and recommended temperatures).
    c. Heat loads at each use point @ maximum load.
    d. Utility requirements including electrical, water, and air.
    e. Pipe and valve sizes.
    f. Instrumentation, safety devices, and control schemes.
    g. Bill of materials.

12. Lube-oil system arrangement drawing including size, rating, and location of all customer connections.

13. Lube-oil component drawings and data including—
    a. Pumps and drivers:
       1. Certified dimensional outline drawing
       2. Cross section and bill of materials
       3. Mechanical seal drawing and bill of materials
       4. Performance curves for centrifugal pumps
       5. Instruction and operating manuals
       6. Completed data forms for pumps and for drivers.
    b. Coolers, filters, and reservoir:
       1. Fabrication drawings
       2. Maximum, minimum, and normal liquid levels in reservoir
       3. Completed data form for cooler(s).
    c. Instrumentation:
       1. Controllers
       2. Switches

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3. Control valves
4. Gages.

d. Priced spare parts list(s) and recommendations.

14. Electrical and instrumentation schematics and bill of materials:
    a. Vibration warning and shutdown limits.
    b. Bearing—temperature warning and shutdown limits.
    c. Lube-oil—temperature warning and shutdown limits.

15. Electrical and instrumentation arrangement drawing(s) and list(s) of connections.

16. Control and trip system data:
    a. Valve-lift sequence on multivalves and final settings.
    b. Pin-setting diagram.
    c. Control- and trip-system drawings.
    d. Control-setting instructions.

17. Governor cross section and setting instructions.

18. Steam flow versus horsepower curves at normal and rated speeds with normal steam conditions.

19. Steam flow versus first-stage pressure curve for multistage machines or versus nozzle-bowl pressure for single-stage machines at normal and rated speed with normal steam.

20. Steam flow versus speed and efficiency curves at normal steam conditions.

21. Steam flow versus thrust bearing load curve.

22. Extraction performance curves.

23. Steam-rate correction factors for curves 18-22 inclusive with off-design steam as follows:
    a. Inlet pressure to maximum and minimum values listed on data sheet in increments agreed at order.
    b. Inlet temperature to maximum and minimum values listed on data sheet in increments agreed at order.
    c. Speed (80-105 percent, 5-percent increments).
    d. Exhaust pressure to maximum and minimum values listed on data sheet in increments agreed at order.

24. Vibration analysis data including—
    a. Number of blades—each wheel.
    b. Number of vanes—each diaphragm.
    c. Number of nozzles —nozzle block, single valve
       —nozzle section, multivalve.
    d. Campbell diagram for each stage.
    e. Goodman diagram for each stage.
    f. Number of teeth on gear-type coupling (when furnished by turbine vendor).

25. Lateral critical analysis report including—
    a. Method used.
    b. Graphic display of bearing and support stiffness and its effect on critical speeds.
    c. Graphic display of rotor response to unbalance (including damping).
    d. Graphic display of overhung moment and its effect on critical speed (including damping).
    e. Journal static loads.
    f. Stiffness and damping coefficients.
    g. Tilting-pad geometry and configuration:
       1. Pad angle
       2. Pivot clearance
       3. Pad clearance
       4. Preload.

26. Allowable flange loading(s) for all customer connections.

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27. A coupling alignment diagram, including recommended limits during operation. Note all shaft-end position changes and support growths from 70 F (21 C) ambient reference.

28. Weld procedures.

29. Hydrostatic test logs.

30. Mechanical run test logs including—
    a. Oil flows, pressures, and temperatures.
    b. Vibration, including X-Y plot of amplitude and phase angle versus revolutions per minute during startup and shutdown.
    c. Bearing metal temperatures.
    d. Observed critical speeds (for flexible rotor).

31. Rotor balance logs.

32. Rotor mechanical and electrical runout.

33. "As-built" data sheets.

34. "As-built" dimensions (including design tolerances) or data:
    a. Shaft or sleeve diameters at—
        1. Thrust collar (for separate collars).
        2. Each seal component.
        3. Each wheel (for "stacked" rotors).
        4. Each interstate labyrinth.
        5. Each journal bearing.
    b. Each wheel bore (for "stacked" rotors) and OD.
    c. Each labyrinth or seal-ring bore.
    d. Thrust-collar bore (for separable collars).
    e. Each journal-bearing ID.
    f. Thrust-bearing concentricity (axial runout).
    g. Metallurgy and heat treatment for—
        1. Shaft.
        2. Wheels.
        3. Thrust collar.
        4. Blades (buckets).

35. Operating and maintenance manuals—manuals shall be furnished describing installation, operation, and maintenance procedures. Each manual shall include the following sections:

    Section 1 — Installation:
        a. Storage
        b. Foundation
        c. Grouting
        d. Setting equipment, rigging procedures, component weights, and lifting diagram
        e. Alignment
        f. Piping recommendations including allowable flange loads
        g. Composite outline drawing for driven/driver train including anchor-bolt locations
        h. Dismantling clearances

    Section 2 — Operation:
        a. Start-up
        b. Normal shutdown
        c. Emergency shutdown
        d. Operating limits
        e. Lube-oil recommendations

    Section 3 — Disassembly and Reassembly Instructions:
        a. Rotor in casing
        b. Rotor unstacking and restacking procedures

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c. Journal bearings for tilting-pad bearings, provide "go-no-go" dimensions with tolerances for three-step plug gages
d. Thrust bearing
e. Seals
f. Thrust collar
g. Wheel reblading procedures

Section 4 — Performance Curves:
a. Steam flow versus horsepower
b. Steam flow versus first-stage pressure
c. Steam flow versus speed and efficiency
d. Steam flow versus thrust-bearing load
e. Extraction curves
f. Steam condition correction factors (prefer monograph)

Section 5 — Vibration Data:
a. Vibration analysis data
b. Lateral critical speed analysis

Section 6 — "As-Built" Data:
a. "As-built" data sheets
b. "As-built" dimension or data (refer to item 34 above)
c. Hydrostatic test logs
d. Mechanical run test logs
e. Rotor balance logs
f. Rotor mechanical and electrical runout at each journal

Section 7 — Drawing and Data Requirements:
a. Certified dimensional outline drawing and list of connections
b. Cross-sectional drawing and bill of materials
c. Rotor drawing and bill of materials
d. Thrust-bearing assembly drawing and bill of materials
e. Journal-bearing assembly drawing and bill of materials
f. Seal-component drawing and bill of materials
g. Lube-oil schematic and bill of materials
h. Lube-oil arrangement drawing and list of connections
i. Lube-oil component drawings and data
j. Electrical and instrumentation schematics and bill of materials
k. Electrical and instrumentation arrangement drawing and list of connections.
l. Control- and trip-system drawings and data
m. Trip- and throttle-valve construction drawing

NOTE: Items 7, 11, 12, 13, 24F, and 35 (Section 7—G through l) inclusive required only for turbine manufacturer's scope of supply.

36. Spare parts recommendation and price list—see 6.2.6 in text of standard.

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